

Honeywell

WebGUS
Planning and Implementation
Guide

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About This Document

This document describes,

- The network requirements to set up and use WebGUS.
- How to install the WebGUS control.
- How to change the WebGUS operating environment.
- How to use WebGUS to view GUS displays from a remote PC using Internet Explorer.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
Display Authoring Tutorial	GU07
Display Scripting User's Guide	GU24
GUS HCI_OPC Data Access User's Guide	GU2B
HCI Client Developer's Guide	TP60
HCI Server Developer's Guide	TP14
TPS System Administration Guide	TP06
TPS Terms and Definitions	TP43
WebGUS Authoring & Runtime Guide	WG20

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Symbol	Definition
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground: Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. Planning for WebGUS

1.1 About the WebGUS Planning and Implementation Guide

Who should use this guide

Individuals should use this publication to:

- learn about network requirements to set up and use WebGUS.
- install the WebGUS control.
- change the WebGUS operating environment.
- learn how to use WebGUS to view GUS displays from a remote PC using Internet Explorer.

Additional references

More information about scripting the WebGUS control and Honeywell servers can be found in:

- *WebGUS Authoring and Runtime Guide.*
- *GUS HCI/OPC Data Access User's Guide.*
- *GUS Display Authoring Tutorial.*
- *GUS Display Scripting User Guide.*
- *TPS System Administration Guide.*
- *TPS Terms and Definitions.*

1.2 WebGUS Functional Overview

What WebGUS provides

The WebGUS product brings operational control room GUS displays to the supervisory desktop PC, without requiring changes to existing libraries of operational GUS displays. It opens the availability of GUS display functionality to any internet-connected computer with the appropriate software configuration and security access rights.

Two primary areas of functionality are provided:

- An ActiveX control encapsulating the GUS runtime functionality.
- Data access redirection, allowing existing GUS display data references to be redirected to one or more surrogate HCI/OPC data servers.

After the WebGUS software is installed on a customer's site server, the product can be downloaded to customer desktops through a Web page-oriented download-and-configure procedure.



WARNING

Do not download the WebGUS package on a GUS node. While the WebGUS ActiveX control is installed with the Display Builder and the Display Runtime, the WebGUS package itself is intended exclusively for desktop nodes, as it provides the capability to redirect LCN data references at runtime. This redirection will not work on a GUS node.

How WebGUS works

WebGUS is an ActiveX control that can host a GUS display. WebGUS also has the ability to redirect GUS display data references to different data servers. WebGUS, therefore, enables a user to view GUS displays, and therefore plant operations, through a desktop browser using a WebGUS ActiveX control within a Web page.

Because GUS displays are also ActiveX containers, a GUS node can leverage WebGUS to encapsulate separate GUS displays into other main GUS displays. This alternative to the existing approach of embedding GUS displays (display source) directly into main displays has potential benefits.



ATTENTION

It is important to note, however, that WebGUS should not be considered interchangeable with embedded displays as described in the GUS Display Builder Guide.

Also, the GUS Display Builder product may now be installed on a desktop. When used in conjunction with the WebGUS software, WebGUS displays can be developed, tested, and executed all within the desktop environment. Such displays may then be made available for subsequent use as WebGUS displays and/or as normal GUS displays.

For example, a plant engineer may use the GUS Display Builder on the desktop to build GUS displays that use only HCI/OPC data references. Or, a plant engineer may develop GUS displays on the desktop, including LCN data references, test them using the WebGUS product's data redirection functionality, and then deploy (after validating against the LCN) them to the control room. After it is deployed to the control room in standard GUS nodes, the LCN references built into the displays work as normal.



WARNING

WebGUS is a read only tool and is currently not intended for writing data to the node using OPC.

WebGUS Download and Configuration Page

The WebGUS Download and Configuration Web page bundles all the runtime settings necessary for execution of GUS displays embedded in Web pages. The bundle of runtime settings constitutes an Environment Definition. The page stores a Current Environment data in scope of the current Windows NT user. WebGUS Environment Settings precede other sources of configuration settings, such as the configuration of HCI Data Access servers in the Domain.

WebGUS definitions



REFERENCE - INTERNAL

See *TPS Terms & Definitions* in Knowledge Builder for WebGUS specific definitions.

1 Planning for WebGUS

1.2 WebGUS Functional Overview

WebGUS Requirements and Resources

System requirements

See the different topology examples in section "Determining WebGUS Resource Topologies" in this guide.

Hardware requirements

Desktop general configuration requirements are consistent with GUS node requirements, which include, as a minimum:

- 450 MHz Pentium III processor
- 128 MB of RAM
- 1.0 GB hard disk space



TIP

Desktops need not be a specifically qualified GUS hardware platform in order to run WebGUS.

Software requirements

As a minimum, the desktop PC that is to use WebGUS requires:

- Microsoft Windows 2000 operating system.
- Microsoft's Internet Explorer (IE) 5.5 or later, for viewing WebGUS Displays within IE Web pages.

Buildtime editing resources

Microsoft standards-compliant ActiveX containers may provide a buildtime editing environment supporting the inclusion and scripting of WebGUS ActiveX controls. Those that are specifically qualified for WebGUS ActiveX controls include the following:

- Microsoft FrontPage 2000
- Honeywell GUS Display Builder

Other components

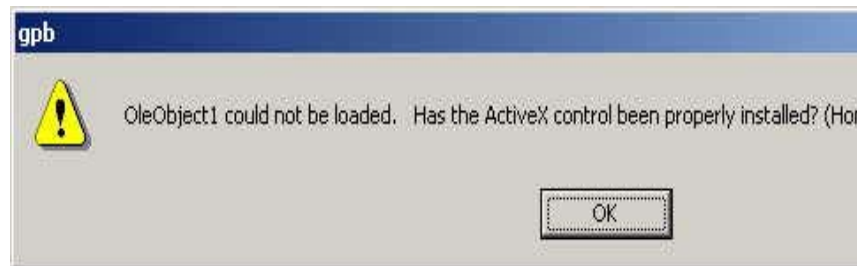
In order to fully operate GUS displays with various other software components that may be incorporated into the displays, the following associated software packages and/or components may also be required:

- Third party ActiveX controls



TIP

GUS displays can still operate, albeit to a lesser extent, if a required ActiveX component is not available to the display. Also, for each existing GUS Display Builder runtime documented behavior, the following error dialog is displayed when a missing ActiveX (in this case named "OleObject1") is encountered at display load time (runtime).



ATTENTION

TPN-integrated Honeywell ActiveX controls including Trend, AlarmWindow and Faceplate are not required and do not operate on the desktop. If they are missing, the WebGUS display emulates the same "red-x" error-condition behavior as they present when used within WebGUS displays, and the above dialog is not presented.

1 Planning for WebGUS

1.2 WebGUS Functional Overview

Nonessential Honeywell components

Unlike standard GUS nodes, WebGUS displays executing on a desktop do not require nor use the *Button Service*, *Picture Manager*, or other services provided by the GUS package. This both simplifies and frees the desktop installation to a simple “download and optional configure” before being able to execute GUS Displays as WebGUS Displays.

It is not necessary for the desktop computer to be in a domain to be able to execute WebGUS displays.

1.3 WebGUS Planning Checklist

Checklist for deploying WebGUS

The following table lists the high-level tasks required to enable users to start using the WebGUS control to view plant operations through GUS displays embedded in a Web browser.

Task	See	Completed
Determine WebGUS configuration: • File Servers • Web Servers • OPC/HCI DA servers • WebGUS Desktops • WebGUS Installation Computer • Windows domains where servers and desktops are located and their relationships • Location of files for Default Environment, WebGUS Distribution Files, Multiple Environments, GUS Displays, Web Pages • WebGUS users or groups • WebGUS user roles		☐
Insure <i>your</i> WebGUS users or groups have adequate permissions or rights to access your Web servers, file servers, and HCI/OPC DA servers. This will involve: • Web server security • HCI security • File server security	Applying security to support WebGUS deployment	☐
Install the WebGUS product from the Honeywell-supplied WebGUS CD to the WebGUS Installation Computer in the Program Files\Honeywell\TPS\WebGUS directory.	See the Software Change Notice (SCN) shipped with the WebGUS software.	☐

1 Planning for WebGUS

1.3 WebGUS Planning Checklist

Task	See	Completed
Prepare for WebGUS desktop download. - Initial local WebGUS Download to the WebGUS Installation Computer. - Create a Default WebGUS Environment.	Using the Environment Configuration Interface" in this guide.	☐
If necessary, modify the CODEBASE URL and DefaultConfigExample.xml file path in the WebGUS Download and Configuration pages (Webgus_download_config.htm and Webgus_env_config.htm).	"How WebGUS Environments Will Function" in this guide	☐
Copy or Publish the WebGUS package including: - The modified Default WebGUS Environment - The modified WebGUS Download and Configuration pages from the WebGUS Installation Computer to their assigned location, based on the selected topology.	WebGUS Administration Help	☐
Test the desktop scenario – download WebGUS control, set current environment, accessing a WebGUS Web page, data access for a GUS display in WebGUS.		☐
Notify your users how they are to download the WebGUS control.		☐
(Optional) Create different environments depending on your user's needs.		☐
(Optional) Inform your users how they can modify and save their Current Environment .		☐

1.4 Important Concepts

WebGUS is a Client

WebGUS is

- A **client** of data servers (for example, TPN Server and PHD Server),
- A **client** of a Web Server that contains the WebGUS distribution files and Sample Web pages, AND
- A **client** of file server(s) that contain folders of GUS displays and WebGUS Environment Definitions.

The need for a web server and file server(s) to support WebGUS is optional. The desktop may contain the resources needed by the WebGUS client.

Security for WebGUS

A WebGUS user must have security defined for all WebGUS resources.

Secured Object	Security Method
HCI Data Servers.	Access to data is controlled by the HCI Security configuration. REFERENCE: Refer the following sections in the <i>TPS System Administration Guide</i> for descriptions of HCI security: • "HCI/OPC Server Capability Files" • "Customizing HCI/OPC Server Capability Files" • "Customizing HCI Server Priority Files"
OPC Data Servers	No inherent security provided by OPC Data Servers.
GUS Displays	Access to GUS Displays is controlled through the normal Windows security mechanisms (that is, assigning permissions to the GUS Display files).

1 Planning for WebGUS

1.4 Important Concepts

Secured Object	Security Method
WebGUS Environment Definitions.	Access to WebGUS Environment Definitions is controlled through the normal Windows security mechanisms (that is, assigning permissions to the files containing WebGUS Environment definitions).
WebGUS Distribution Files	Access to WebGUS Distribution Files is controlled through the normal Windows security mechanisms (that is, assigning permissions to the WebGUS Distribution Files).
WebGUS Web pages	Access to WebGUS Web pages is controlled through the normal Windows security mechanisms (that is, assigning permissions to the WebGUS Distribution Files).

2. Getting Started

2.1 WebGUS Packages on the Product CD

Installation of any of the three WebGUS packages creates a directory under <specified drive>:\Program Files\Honeywell\TPS with the same name as the package.

Honeywell provides three installable packages on the distribution CD.

WebGUS Distribution Package

This package contains files to enable WebGUS to function on the Web.

File	Description
WebGUS Cabinet (CAB)	Downloadable code for the WebGUS ActiveX control and Redirection Component
WebGUS Download and Configuration HTM	A template Web page containing a reference to the CAB file location for the download source and a reference to the Default Environment XML file location for the default environment source. The initial access to this page both downloads the WebGUS control and initializes a default environment.
Default Environment XML	A template environment which provides the format for any subsequent environment definitions.
WebGUS Environment Configuration HTM	A template Web page for initializing and manipulating the WebGUS environments. This page does not download the WebGUS control.
Industrial Theme Background JPG	A JPG providing the textured background for the Web pages.

2 Getting Started

2.1 WebGUS Packages on the Product CD

WebRAC Distribution Package

This package contains sample template and help files.

File	Description
WebGUS Download and Configuration HTM, Default Environment XML, WebGUS Environment Configuration HTM.	Backup copies of WebGUS installation files
WebGUS Samples	Reusable template HTM pages and PCT displays to be used "out of the box" to allow a customer site to get a quick start on using WebGUS

LCN Redirection Distribution Package

This package contains the files for the standard LCN Redirection component, a customizable component for redirecting LCN data access.

File	Description
WebGUS Data Visual Basic Project	Files for the Standard Honeywell implementation for LCN Redirection
LCN Redirection Word Document	Documentation describing the Honeywell implementation and how to customize it

Determine the WebGUS Overall Topology for Your Plant

Before installing and setting up WebGUS, prepare for the installation and setup by gathering information and making decisions about the local WebGUS resource topology.

At a minimum, to run GUS displays, WebGUS requires the following:

- A desktop computer
- An OPC Data Access Server

WebGUS is capable of communicating with a number of HCI/OPC Servers and there are many WebGUS resource topologies from which to choose. It is impossible to enumerate all the possible configuration combinations.

This document describes two example scenarios to use as a sample guide through the setup process. The first step for either scenario is to assign a networked computer to be the WebGUS Installation Computer.

The WebGUS Installation Computer can be any Windows 2000 or Windows XP computer not connected to the TPN. An Administrator uses the WebGUS Installation Computer to do the following:

- Install the WebGUS product CD
- Prepare WebGUS for desktop download

"Scenario 1" describes a simple topology. A networked computer serves as both the Installation Computer and the desktop PC. All resources (for example, GUS displays, web pages, and WebGUS environment definition files) reside on this computer. The data access server is in the same domain as the Desktop PC.

"Scenario 2" describes a more complex topology. Resources are distributed among a web server, file server, Installation Computer, and desktop PC. The data access server is in a different domain than the desktop PCs.

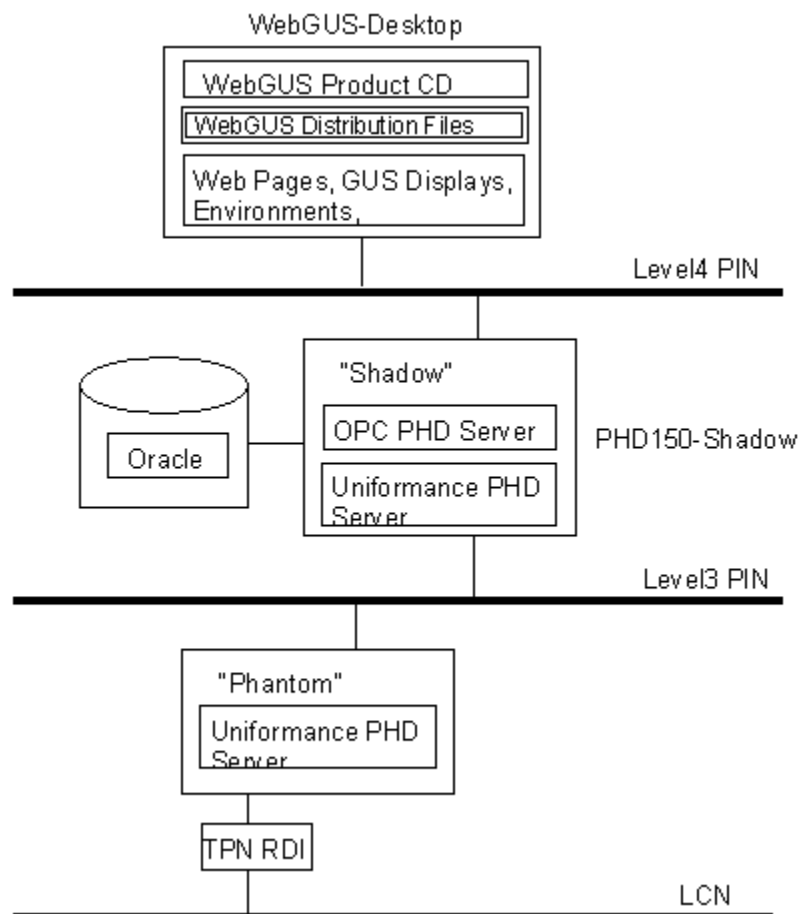
2 Getting Started

2.2 Implementation Scenario 1 – A Simple Topology

2.2 Implementation Scenario 1 – A Simple Topology

Topology Overview

Below is an example WebGUS topology.



Example site configuration

- WebGUS_Desktop and PHD150_Shadow are in the same Windows Domain.
- WebGUS_Desktop serves as the WebGUS Installation Computer and WebGUS Desktop PC.
- All GUS Displays reside locally in the c:\picture directory on the WebGUS_Desktop.
- All web pages and environment files reside locally on the WebGUS_Desktop.

Determine and Implement Security Model for Your Plant in Support of WebGUS

In the example, the WebGUS_Desktop and PHD150_Shadow are in the Windows domain called Domain1 with *user1* as the WebGUS user domain account. The OPC PHD Server DCOM settings and PHD150_Shadow User Rights must be configured properly for the WebGUS user to access the PHD150 data. The configuration is as follows:

- WebGUS user “user1” must have proper access and launch user rights to the OPC PHD Server.

Normally, the default installation configuration of the OPC PHD Server provides the appropriate rights.

- WebGUS user “user1” must be able to access PHD150_Shadow from the network.

Use the following procedure to, setup the “Access This Computer From The Network” User Rights.

Step	Action
1	Start > Programs > Administrative Tools (Common) > User Manager for Domains > Policies > User Rights.
2	Select “Access this computer from the network” from the dropdown list, add Domain\user1 to the access list if necessary. Normally, the default Windows NT security settings satisfies your needs.



REFERENCE

Refer to the “Configuring the OPC PHD Server” section in the *Uniformance OPC PHD Server User’s Guide* for additional information

2 Getting Started

2.2 Implementation Scenario 1 – A Simple Topology

Install the WebGUS on the WebGUS Installation Computer

Preconditions

To install the WebGUS package on the Installation Computer (also the WebGUS_Desktop in this scenario):

- The WebGUS_Desktop is a PC running Windows 2000 and connected to a LAN.
- IE 5.5 or later is installed on the WebGUS_Desktop.



ATTENTION

Requirements for WebGUS PCs:

- Install the **System Management Runtime** component.
 - Install the **WorkStation Security** package.
 - Execute the script **Linkdomaingroups.vbs**.
-

Procedure to install WebGUS on Installation Computer

Step	Action
1	Log onto the WebGUS_Desktop as an account with local administrator privilege.
2	Insert the TPS System Software CD.
3	Select the WebGUS package to install.
4	Select WebGUS components, WebGUS and WebRAC, to install on the WebGUS_Desktop. Note: You only need to install the LCN Redirection component if you plan to modify the source code to implement your plant's own LCN Redirection Algorithm.

Perform Initial WebGUS Desktop Download

The WebGUS installation process copies all the files in the packages to the directory <specified drive>:\Program Files\Honeywell\TPS.

For the desktop users to run GUS display using WebGUS, the webgus.cab file has to be downloaded on the desktop computer in the directory C:\WINNT\Downloaded Program Files. The downloading process will download the webgus.cab and register the components using **Internet Component Download** technology.

After the WebGUS downloading, the component “Honeywell WebGUS Control V310” will be shown in the directory C:\WINNT\Downloaded Program Files

Preconditions:

- The WebGUS and WebRAC packages are installed on the WebGUS Installation Computer (WebGUS_Desktop).

Follow the steps in the following procedure.

Step	Action
1	Logon as an account with local administrator privilege on the WebGUS_Desktop.
2	Double click on <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebGUS\webgus_download_config.htm to download WebGUS locally. WebGUS is downloaded to the WebGUS_Desktop and the default environment is recorded in the Registry. The WebGUS page shows the default environment settings in IE.

2 Getting Started

2.2 Implementation Scenario 1 – A Simple Topology

Customize the WebGUS Default Environment

Preconditions:

- HCI/OPC Data Access servers and LCN Data Redirection information is available

On the WebGUS Download and Configuration Page, do the following procedure

Step	Action
1	Edit the GUS Display Search Path. In the simple scenario example, the GUS displays will be located locally in the directory c:\Pictures on the WebGUS_Desktop
2	Edit the HCI/OPC DA Servers Information. In the example: HCI/OPC Data Access Servers: Name: PHD150 ProgID: hci.PHD Host: PHD150_Shadow Description and Collection Rate are optional. Refer to "Administering for HCI/OPC Data Access" in this guide for details.
3	Edit LCN Data Redirection information. In the example: LCN Data Redirection: All points using LCN name form in the GUS displays will be redirected to PHD150 Refer to the " Setting Environment for LCN Data Redirection" section in this guide for additional information.
4	Click Save As Environment to overwrite <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebGUS\DefaultEnvironment.xml on the WebGUS_Desktop. Note: Do not change Runtime Timeout Settings at this step. Refer to "Modifying an Environment Setting" in this guide for more information.
5	Click "Set As Current Environment" to apply the new default environment settings to the WebGUS_Desktop Registry. Refer to "How WebGUS Environments Will Function" in this guide for details.

The modified DefaultEnvironment.xml appears in IE as shown below.

WebGUS Download And Configuration

Buttons: Browse for Environment..., Save Environment As..., Set As Current Environment, Show Current Environment, Help

Environment Settings

GUS Display Search Path:

OPC / OPC Data Access Servers

Defines, supplements, or overrides server definition by TPS domain.

Name	Prog ID	Host Computer	Description	Collection Rate(sec)	Ping Parameter
FHCD150	FHCD150	FHCD150_Shadow	FHCD150 Shadow Server		

ON Data Redirection

Rename Point:

Insert Prefix:

Redirect Point:

Point Prefix	OPC / OPC Surrogate Server
F	FHCD150

2 Getting Started

2.2 Implementation Scenario 1 – A Simple Topology

View GUS Displays in Sample Web Page

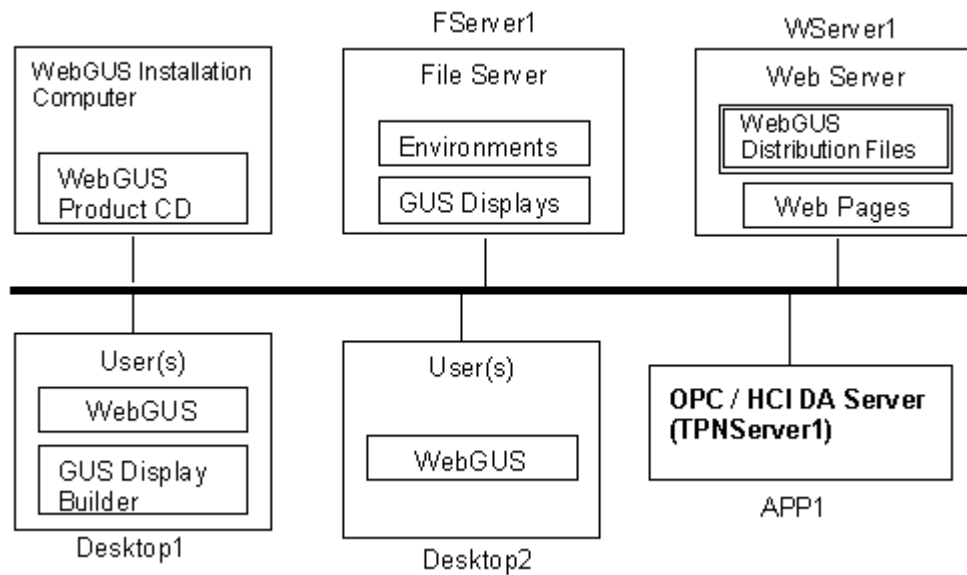
Do the following procedure to view the sample web page.

Step	Action
1	Logon as a WebGUS user user1 on the WebGUS_Desktop.
2	On the WebGUS_Desktop, navigate to the sample html file <specified drive>:\Program Files\Honeywell\TPS \WebRAC\Samples\WebSamples\selectedgusdisplay.htm,. Execute the page, and load your GUS display.
3	Verify the GUS display shows process data correctly

2.3 Implementation Scenario 2 -- A Complex Topology

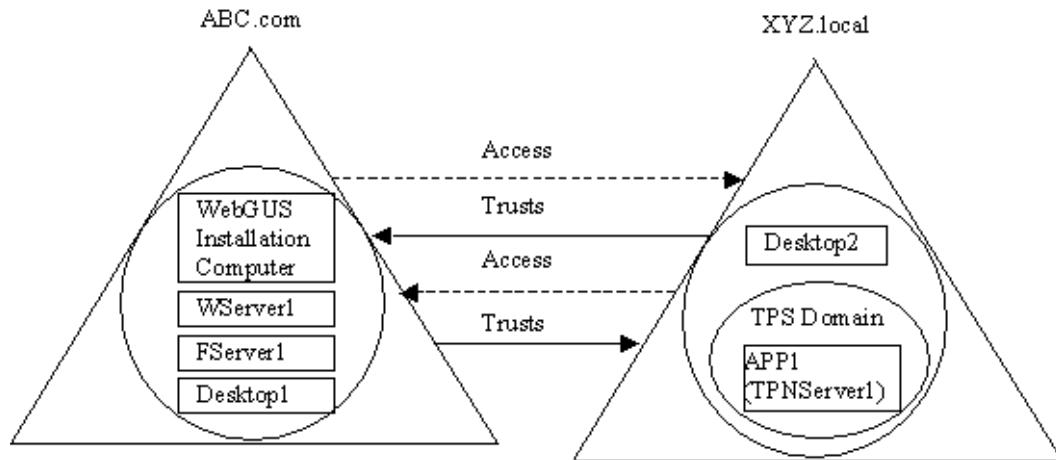
Topology overview

Below is an example WebGUS topology. This is a variation of the Plant topology shown in this guide.



2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology



The example site has the following configuration:

- Domain XYZ.local and domain ABC.com are in the different Windows 2000 or Windows 2003 forests.
- There are two network shares named WebGUS_Environments and GUS_Displays on the file server (FServer1). The share WebGUS_Environments will be used to store multiple WebGUS environment definitions and the share GUS_Displays will be used to store the GUS displays.
- OPC/HCI Data Access Server: APP1 is an APP Node with the TPN Server software package installed.
- In the ABC.com domain, the following accounts will be used as the WebGUS desktop users with different permissions.

Fred is the author for the WebGUS pages and GUS displays, he needs to access the process data from the desktop computer as a view only user.



TIP

Fred may also be running a TPN connected GUS node to perform full validation of the GUS displays.

Mary is the plant manager who needs to access the process data from the desktop computer as a view only user.

User	WebGUS Pages	GUS_Displays	TPNServer1
Fred	Read/Write	Read/Write	Local View Only Users Permission
Mary	Read	Read	Local View Only Users Permission

- In the XYZ.local domain, the following accounts will be used as the WebGUS desktop users with different permissions.

Joe needs to access the process data from the desktop computer as an engineer.

User	WebGUS Pages	GUS_Displays	TPNServer1
Joe	Read	Read	Local Engineers Permission

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Setup for Complex Topology

Determine and Implement Security Model for Your Plant in Support of WebGUS

Each site has different WebGUS topology and security requirements. The following setup procedure explains the security settings necessary for the example. Each individual site has to determine its security settings based on the topology and security requirements.

The basic requirement of WebGUS security is that any server application that provides access to a protected resource (for example, a GUS display file or an HCI/OPC data access server) must be able to positively identify the users that are requesting services.

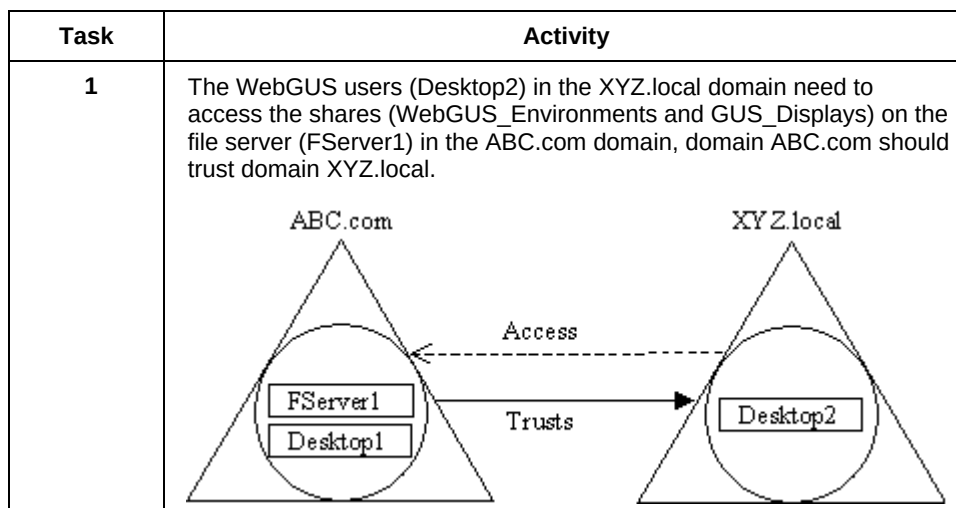
The authentication and authorization process is performed by various system infrastructure components, within the Windows 2000 or Windows 2003 user domain structure.

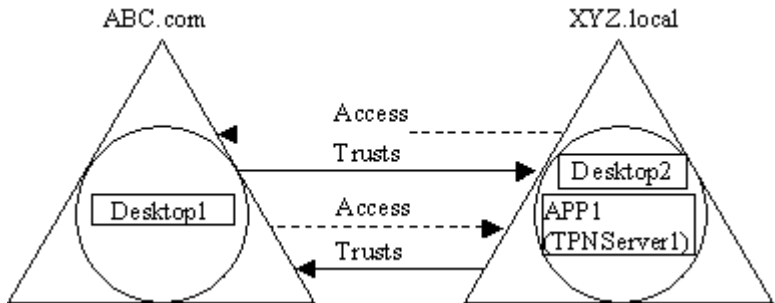
In the example, all the following security settings have to be properly configured in order for the WebGUS desktop users to view the GUS displays.

Refer to the "WebGUS Security requirement" section in this guide for more information.

Identify Trust Relationships Among Domains

Do the following in order to ensure that the proper trusts are established between Domains.



Task	Activity
2	The WebGUS users (Desktop1) in the ABC.com domain need to access the shares (WebGUS_Environments and GUS_Displays) on the file server (FServer1) that resides in the same domain. There are no authentication issues.
3	The WebGUS users (Desktop2) in the XYZ.local domain need to access the TPN Server (TPNServer1) that resides in the same domain. There are no authentication issues.
4	The WebGUS users (Desktop1) in the ABC.com domain need to access the TPN Server (TPNServer1) in the XYZ.local domain, domain XYZ.local should trust domain ITDomain.com. The TPN Server (TPNServer1) in the XYZ.local domain needs to callback the client (Desktop1) on the data change events, domain ABC.com should trust domain XYZ.local as well. 

Refer to Microsoft Documents for more detailed information about Domain Trust Relationships.



ATTENTION

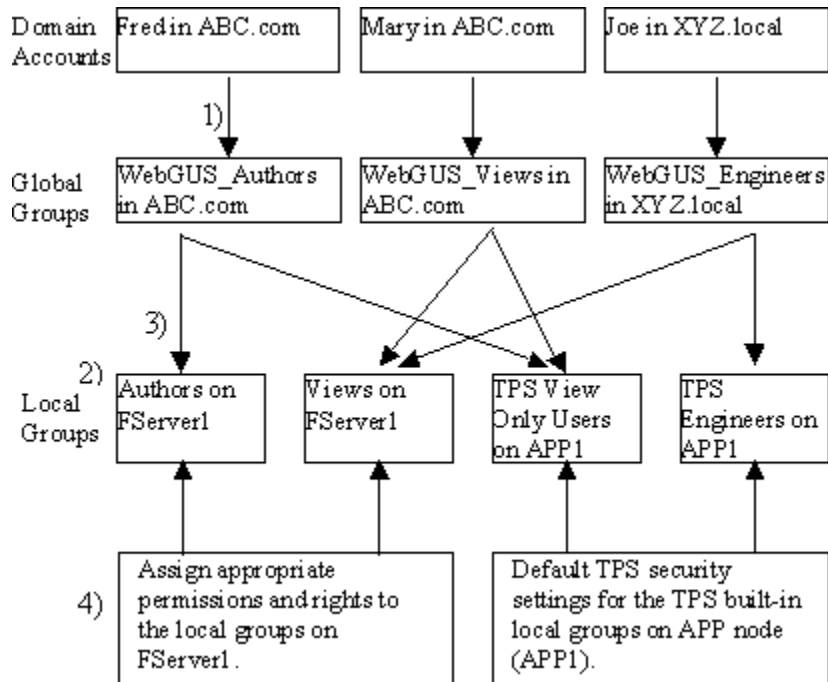
Currently, the HCI-based Data Access servers require a two-way trust relationship between the client and the server domains.

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Develop a Group Strategy

The guidelines for implementing global group and local group are:



Step	Action
1	Place user accounts with common job responsibilities into a global group.
2	Create a local group for a group of resources to be shared in common.
3	Place the global groups into the appropriate local groups.
4	Assign permissions to the local group.

Refer to Microsoft Documents for more detailed information about Group Strategies.

Implement Security Settings to Support the Group Strategy***File Server Security***

The following security settings need to be verified or configured on the file server (FServer1) or on the Domain Controller in the ITDomain.com.

Perform the following procedure on the Domain Controller in the ABC.com domain.

Step	Action
1	Log onto the Domain Controller in the ABC.com as the domain Administrator.
2	Create the global groups WebGUS_Authors and WebGUS_Views.
3	Add Fred, Mary to WebGUS_Authors and WebGUS_Views global groups respectively.

Perform the following procedure on the File Server (Fserver1) in the ABC.com domain.

Step	Action
1	Log onto the file server (FServer1) as the local Administrator.
2	Create local groups Authors and Views for the network resources WebGUS_Environments and GUS_Displays to be shared.
3	Add domain global groups WebGUS_Authors and WebGUS_Views to the local group Authors and Views respectively. Note: After the global group WebGUS_Engineers is created in the XYZ.local, add domain global group WebGUS_Engineers to the local group Views.
4	Grant the local group Authors RW permissions to the network shares WebGUS_Environments and GUS_Displays.
5	Grant the local group Views RX permissions to the network shares WebGUS_Environments and GUS_Displays. This will automatically add Read & Execute, List Folder Contents and Read Permissions to the local group Views.

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Step	Action
6	Apply the User Rights "Access This Computer From the Network" to the AUTHENTICATED USERS group: a) Select the following from the Task Bar: Control Panel > Administrative Tools > Local Security Policy b) Select Local Policies > User Rights Assignment. c) Click Access This Computer From Network. d) Add AUTHENTICATED USERS.

Web Server Security

Use your local procedures to install and setup the web server. The web server set up procedure is performed using the Internet Services Manager Snap-In.

Verify Web Server security has been setup properly for the WebGUS users from the ABC.com and XYZ.local domain to download WebGUS control and to access web pages.

Refer to Microsoft Documents for more detailed information about Web Server installation and setup.

Firewall exceptions

Manually add the Web Server service to the Windows Firewall exceptions on the web server:

Step	Action
1	At the WebGUS Web Server, select Start>Control Panel> Security Center>Windows Firewall .
2	Select Advanced tab.
3	Select an appropriate network and click Settings under Network Connection Settings .
4	In Advanced Settings , select Web Server .

HCI/OPC Data Access Server Security

The following security settings need to be verified or configured on the APP Node (APP1) or the Domain Controller for the WebGUS desktop users to access HCI/OPC Data Access Server (TPNServer1).

Perform the following procedure on the Domain Controller in the XYZ.local domain.

Step	Action
1	Log onto the Domain Controller in the XYZ.local as the domain Administrator.
2	Create a global group WebGUS_Engineers.
3	Add Joe to WebGUS_Engineers global group.

Perform the following procedure on the APP Node in the XYZ.local domain.

Step	Action
1	Log onto the APP Node (APP1) as the local Administrator
2	- Add domain global groups WebGUS_Authors and WebGUS_Views to the TPS built-in local groups Local Engineers and Local View Only Users, respectively. - Add domain global group WebGUS_Engineers to the local group Local Engineers.

Note: Default TPS security settings are used for the TPS built-in local groups

**REFERENCE**

Refer to the "TPS Security Customization" section in the *TPS System Administration Guide* for additional information.

- | | |
|---|--|
| 3 | Set the HCI Server's method security to permit access to the server's secure methods (Read, Write, and Shutdown, and so forth). <ul style="list-style-type: none"> The capability file security setting will be done on each node in the TPS domain OU. The default TPS security settings will be used. The WebGUS users or groups will have the permissions of the TPS built-in local groups for access to the HCI/OPC Data Access Server . |
|---|--|

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Step	Action
	REFERENCE Refer to the "Customizing HCI/OPC Server Capability Files" section in the <i>TPS System Administration Guide</i> for additional information. Refer to "Accessing HCI Server from Remote Client" in the <i>TPS System Implementation Guide</i> for more information. Refer to "Configuring the TPN Server" in the <i>TPN Server User's Guide</i> for more information.

WebGUS Desktop Security

The following security settings need to be verified or configured on the WebGUS Desktop (desktop1 and desktop2) for the HCI/OPC Data Access Server (TPNServer1) to callback the WebGUS desktops on the data change events.

Perform the following procedure on the the Desktop Computer.

Step	Action
1	Log onto the Desktop Computer as the local Administrator
2	Apply the User Rights "Access This Computer From the Network" to the AUTHENTICATED USERS group. a) Select the following from the Task Bar: Control Panel -> Administrative Tools ->Local Security Policy. b) Select Local Policies -> User Rights Assignment. c) Click "Access This Computer From Network". d) Add "AUTHENTICATED USERS".

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Install the WebGUS on the WebGUS Installation Computer

Preconditions

To install the WebGUS package on the Installation Computer (also the WebGUS_Desktop in this scenario):

- The WebGUS_Desktop is a PC running Windows 2000 and connected to a LAN.
- IE 5.5 or later is installed on the WebGUS_Desktop.



ATTENTION

Requirements for WebGUS PCs:

- Install the **System Management Runtime** component.
 - Install the **WorkStation Security** package.
 - Execute the script **Linkdomaingroups.vbs**.
-

Procedure to install WebGUS on Installation Computer

Step	Action
1	Log onto the WebGUS Installation Computer as an account with local administrator privilege.
2	Insert the TPS System Software CD.
3	Select the WebGUS package to install.
4	Select components WebGUS and WebRAC . Note: You only need to install the LCN Redirection component if you plan on modifying the source code to implement your plant's own LCN Redirection Algorithm.

Perform Initial WebGUS Desktop Download

The WebGUS installation process copies all the files in the packages to the directory <specified drive>:\Program Files\Honeywell\TPS.

For the desktop users to run GUS display using WebGUS, the webgus.cab file has to be downloaded on the desktop computer in the directory C:\WINNT\Downloaded Program Files. The downloading process will download the webgus.cab and register the components using **Internet Component Download** technology.

After the WebGUS downloading, the component “Honeywell WebGUS Control Vxxx” will be shown in the directory C:\WINNT\Downloaded Program Files where xxx is the WebGUS version.

Preconditions:

- The WebGUS and WebRAC packages are installed on the WebGUS Installation Computer.

Perform the following procedure.

Step	Action
1	Logon as an account with local administrator privilege on the WebGUS Installation Computer.
2	Double click on <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebGUS\webgus_download_config.htm to download WebGUS locally. WebGUS is downloaded to the WebGUS Installation Computer and the default environment is recorded in the Registry and the WebGUS page shows the default environment settings in IE.

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Customize the WebGUS Default Environment

Preconditions:

- HCI/OPC Data Access servers and LCN Data Redirection information is available.

Perform the following procedure on the WebGUS Download and Configuration Page.

Step	Action
1	Edit the GUS Display Search Path. In the example, the GUS displays will be located locally in the directory C:\Pictures on the WebGUS desktops and in the network share directory GUS_Displays on the file server FServer1.
2	Edit the HCI/OPC DA Servers Information. In the example: HCI/OPC Data Access Servers. • Name: TPNServer1 • ProgID: hci.TPNServer • Host: APP1 • Description and Collection Rate are optional Refer to "HCI/OPC Data Access Servers" in this guide for more details.
3	Edit LCN Data Redirection information. In the example: LCN Data Redirection. All points using LCN name form in the GUS displays will be redirected to TPNServer1. Refer to "Setting Environment for LCN Data Redirection" in this guide for details.
4	Click "Save As Environment" button to overwrite <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebGUS\DefaultEnvironment.xml on the WebGUS Installation Computer. Note: Do not change Runtime Timeout Settings at this step. Refer to "Modifying an Environment Setting" in this guide for details.
5	Click "Set As Current Environment" to apply the new default environment settings to the Installation Computer Registry. Refer to the "How WebGUS Environments Will Function" section in this guide for additional information.

The modified DefaultEnvironment.xml is shown in IE as below.

WebGUS Download And Configuration - Microsoft Internet Explorer

Address: C:\Program Files\Honeywell\TPS\WebGUS\webgus_download_config.htm

Honeywell WebGUS Download And Configuration WebGUS

Browse for Environment... Save Environment As... Set As Current Environment Show Current Environment

Help

Environment Settings

GUS Display Search Path: C:\pictures\FServer1\GUS_Displays

HCI / OPC Data Access Servers

Defines, supplements, or overrides server definition by TPS domain.

Name	Prog ID	Host Computer	Description	Collection Rate(sec)
TPNServer1	Hci.TPNServer	APP1	TPN Server for Unit 1	

LCN Data Redirection

Rename Point

Insert Prefix:

Redirect Point

Point Prefix	HCI / OPC Surrogate Server
*	TPNServer1

Runtime Timeout Settings:

(Note: Changes to timeout settings may impact the way the system operates and should only be made by an expert user.)

Done My Computer

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

Copy WebGUS Components to the Web Server or File Server

In the example, the WebGUS distribution files will be located in the directory C:\inetpub\wwwroot directory on the Web Server (Wserver1).

Preconditions;

- Web Server has been installed and setup to serve the web content. (In the example, Microsoft IIS has been installed and setup to serve the web content).

Based on this example, perform the following procedure.

Step	Action
1	Copy the WebGUS distribution files from <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebGUS directory on the WebGUS Installation Computer to the web home directory c:\inetpub\wwwroot on the web server (Wserver1). The WebGUS distribution files are consist of: • WebGUS.cab. • Webgus_download_config.htm. • Webgus_env_config.htm • Modified DefaultEnvironment.xml • Indtextb.jpg Note: Webgus_download_config.htm and Webgus_env_config.htm are identical except Webgus_download_config.htm contains CODEBASE attribute in the <OBJECT> tag. Webgus_download_config.htm is used for downloading WebGUS and configuring WebGUS environment settings. Webgus_env_config.htm is used for configuring WebGUS environment settings only.
2	Copy the modified DefaultEnvironment.xml to the share directory WebGUS_Environments on the file server (FServer1). Note: WebGUS desktop users need to be able to browse the DefaultEnvironment.xml on the file server (FServer1). If DefaultEnvironment.xml is stored on the Web Server (Wserver1) only, the WebGUS desktop user will not be able to use the "Browse for Environment" button to browse DefaultEnvironment.xml on the web server (Wserver1) after the WebGUS is downloaded to the desktop.

Step	Action
3	Copy the following sample html files from WebRAC package from <WebGUS Installation Drive>:\Program Files\Honeywell\TPS\WebRAC directory on the WebGUS Installation Computer to the web home directory C:\inetpub\wwwroot on the web server (Wserver1). - Positionwithcategory.htm. - Selectedgusdisplay.htm. - Simplecontainment.htm
4	Copy the GUS displays to the shared directory GUS_Displays as necessary.

Download and Configure WebGUS to the Desktop

Perform the following procedure.

Step	Action
1	Logon as an account with local administrative privilege on the WebGUS desktop computer (Desktop1).
2	Verify the IE security is set properly for download on the desktop1. Medium is default.
3	Navigate to Error! Hyperlink reference not valid..
4	The WebGUS control is downloaded and the default environment is shown on the page in IE.
5	Logoff/Logon as a WebGUS user Fred on the desktop1.
6	Navigate to Error! Hyperlink reference not valid..
7	Verify the default environment setting display on the environment page.
8	Repeat Step 1-7 on the desktop2 using the WebGUS user Joe.

2 Getting Started

2.3 Implementation Scenario 2 -- A Complex Topology

View GUS Displays in Sample Web Page

Perform the following procedure

Step	Action
1	Logon as a WebGUS user Fred on the desktop1.
2	On the desktop1, navigate to the sample html file <specified drive>:\ Program Files\ Honeywell\ TPS\ WebRAC\ Samples\ WebSamples\ selectedgusdisplay.htm. Execute the page, and load your GUS display.
3	Verify the GUS display shows process data correctly.
4	Repeat steps 1-3 on the desktop2 using the WebGUS user Joe.

Verify Security Model Is Implemented Correctly and Is Effective

Perform the following procedure

Step	Action
1	Log onto the desktop1 as Fred.
2	Map the network drives X and Y to WebGUS_Environments and GUS_Displays shares respectively.
3	Verify Fred should be able to Read and Write the files in the drive X and Y.
4	Navigate to the sample html file and load your GUS display.
5	Verify Fred can only view the process data.
6	Log onto the desktop1 as Mary. Repeat step 2-4. Verify Mary can only Read the files in the drive X and Y. Verify Mary can only view the process data.
7	Log onto the desktop2 as Joe. Repeat step 2-4. Verify that Joe can only Read the files in the drive X and Y. Verify that Joe can Read/Write to the process data as a TPS engineer.

2.4 Troubleshoot the Installation of WebGUS

Symptom – Red X Error Indicator

To indicate an ActiveX error, IE displays a red X instead of the WebGUS logo on the WebGUS Download and Configuration Page, as shown in the following screen example.

When a user on the WebGUS desktop computer opens the Internet Explorer (IE) browser and navigates to `http://your web server url/webgus_download_config.htm`, IE prompts the user with a Security Warning Message: “Do you want to install and run `http://your web server url /WebGUS.cab` signed on Date/Time and distributed by: Honeywell Inc. ...”. After the user clicks YES, IE displays the red X.

Error! Objects cannot be created from editing field codes.

Causes of Red X Error Indicator

The following are the most common causes of the red X indicator.

- The version of Internet Explorer is incorrect. WebGUS requires IE5.5 or higher to view WebGUS displays within IE web pages.
- In Windows 2000 Professional/Server, Windows XP, or Windows 2003, members of the Users default group do not have the required permissions, so the Users default group cannot download ActiveX controls.

The download and configuration of WebGUS components modifies the following hives in the local computer registry:

- HKEY_CURRENT_USER – stores WebGUS environment configuration data.
- HKEY_LOCAL_MACHINE – stores registration information of ActiveX control(s) and COM objects that are included with WebGUS.

The download and configuration of WebGUS components writes to the following directories:

- C:\WINNT\Downloaded Program Files
- C:\WINNT\System32
- C:\WINDOWS\Downloaded Program Files
- C:\WINDOWS\System32

ACTION: Make sure you logon as an account with local administrative privilege to download WebGUS.

2 Getting Started

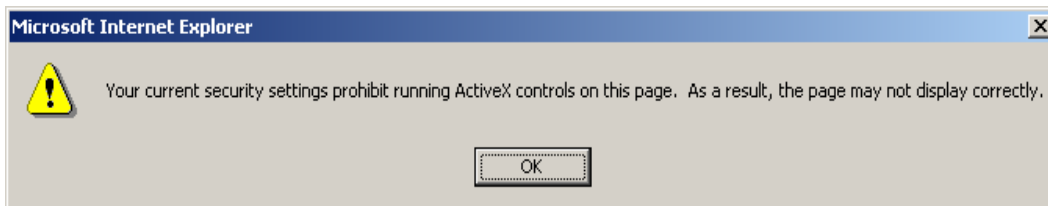
2.4 Troubleshoot the Installation of WebGUS

- If your WebGUS desktop PC has a previous version of the WebGUS control installed, you might not be able to properly download the newer version of the WebGUS.
ACTION: Do the following to remove the older version of the “Honeywell WebGUS Control Vxxx” from the “Downloaded Program Files” folder, and then download a new version of the WebGUS control: Go to C:\WINDOWS\Downloaded Program Files, right-click “Honeywell WebGUS Control Vxxx”, and then click “Remove”.
- Your web server may not have the proper NTFS permissions applied to it. If you choose to use “Allow Anonymous Access”, then the IUSR_<computername> account should have READ (RX) access to the home directory, such as C:\inetpub\wwwroot.
ACTION: If you choose to use the combination of Web Server security and NTFS security, make sure the WebGUS desktop users have the proper permissions to the Web Server and NTFS file system.
- Anti-virus software may interfere with the download of ActiveX controls.
ACTION: To prevent this problem, stop the anti-virus software, download the WebGUS control, and then restart the anti-virus software.

Security error message displayed

A user on the WebGUS desktop computer with the Internet Explorer browser open navigates to the [http:// your web server url/webgus_download_config.htm](http://your_web_server_url/webgus_download_config.htm).

The IE displays the following error message.



In the Internet Explorer browser make sure that your “Security Settings” are not too restrictive to allow the download and installation of the ActiveX controls. Refer to the Internet Explorer manual for additional security setting information. To access the Security Setting in the Internet Explorer, select Tools menu -> Internet Options -> Security Tab.

In addition to the above troubleshooting tips, event log is a very important resource for troubleshooting. The event logs can be viewed by using Event Viewer.

3. WebGUS Resource Topologies

3.1 Determining WebGUS Resource Topologies

Type of WebGUS resources

The resources used by WebGUS consist of:

- WebGUS distribution files from the Honeywell TPS System Software distribution CD.
- GUS displays (*.PCT).
- Web pages that contain WebGUS ActiveX control (*.HTM).
- WebGUS environment configuration files (*.XML).
- OPC/HCI DA servers.

The WebGUS distribution files consist of a single CAB file that contains:

- The WebGUS ActiveX control
- GUS display runtime
- HCI runtime
- LCN Data Redirection Component executable modules

These files will be downloaded to the WebGUS user computers with the help of the WebGUS Download and Configuration page. The WebGUS distribution also contains:

- Web page examples
- WebGUS Download and Configuration page templates
- Environment definition templates
- WebGUS help files
- Other supporting files

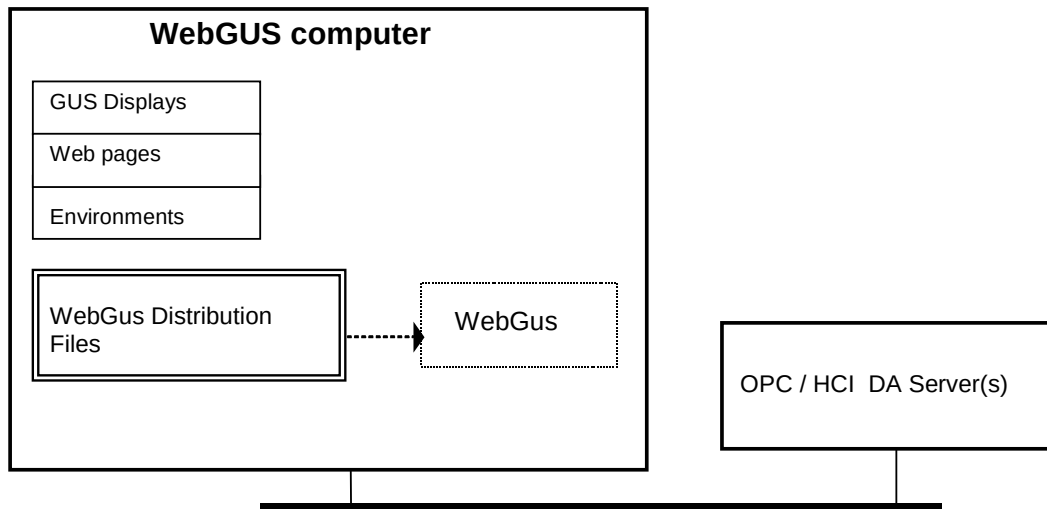
3 WebGUS Resource Topologies

3.1 Determining WebGUS Resource Topologies

Local user topology

The following topology illustrates where WebGUS resources reside on the local computer. All resources are accessible through the file system. The OPC/HCI Data Access servers may be local or are accessible over Ethernet LAN.

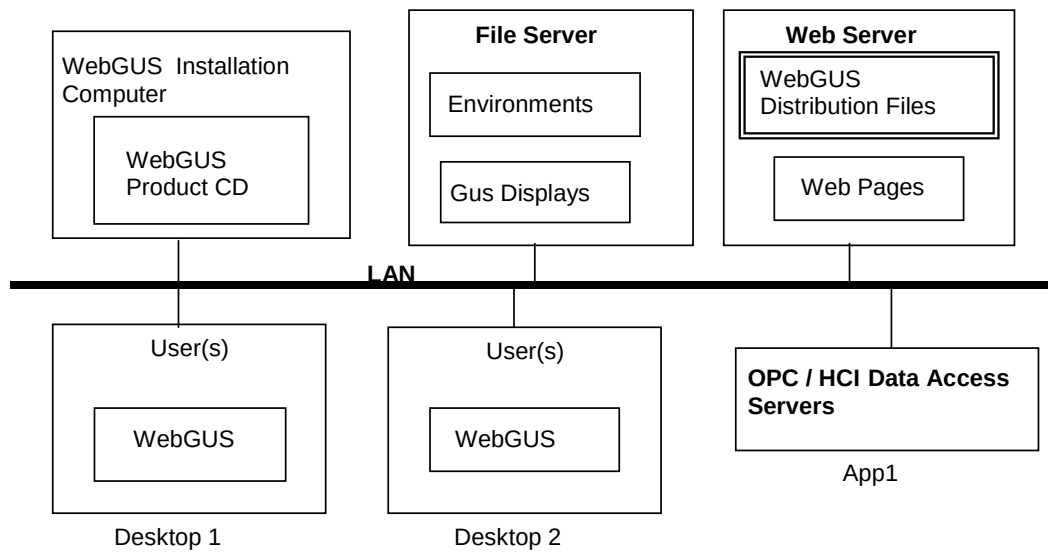
Figure 1 Local User Topology



Plant topology

In the Plant topology, WebGUS resources may reside on multiple computers. The local intranet Web server processes Web pages. File servers and the Web server may be physically located on the same computer.

Figure 2 Plant Topology



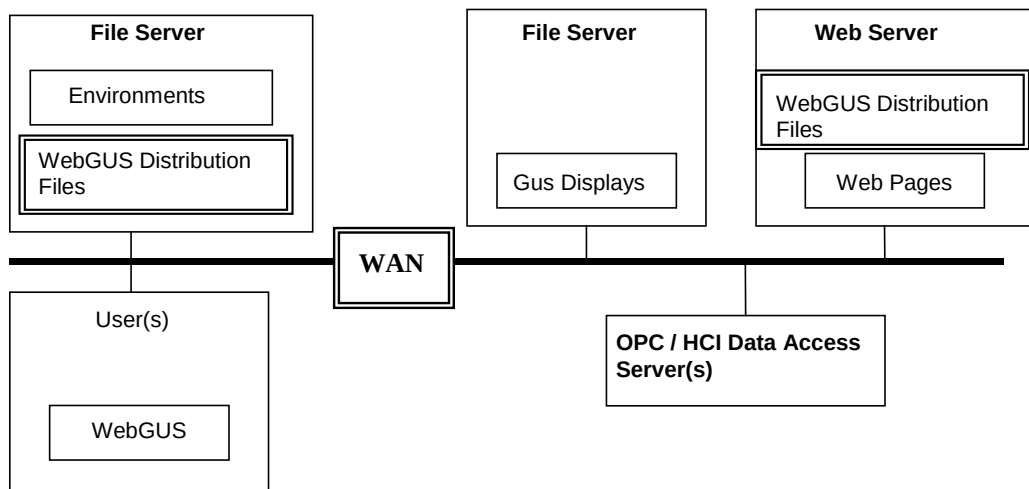
3 WebGUS Resource Topologies

3.1 Determining WebGUS Resource Topologies

Remote plant topology

In this topology, WebGUS users access remote plant's GUS displays and data servers over WAN.

Figure 3 Remote Plant Topology



The configuration of a WebGUS environment file allows specifying remote locations for GUS displays and OPC/HCI Data Access servers. A local desktop user must be able to access remote plant Web pages.



ATTENTION

In this configuration security issues are very significant. The remote plant must be able to authenticate the WebGUS user; additionally, COM communication ports may not be blocked at the intermediate routers.



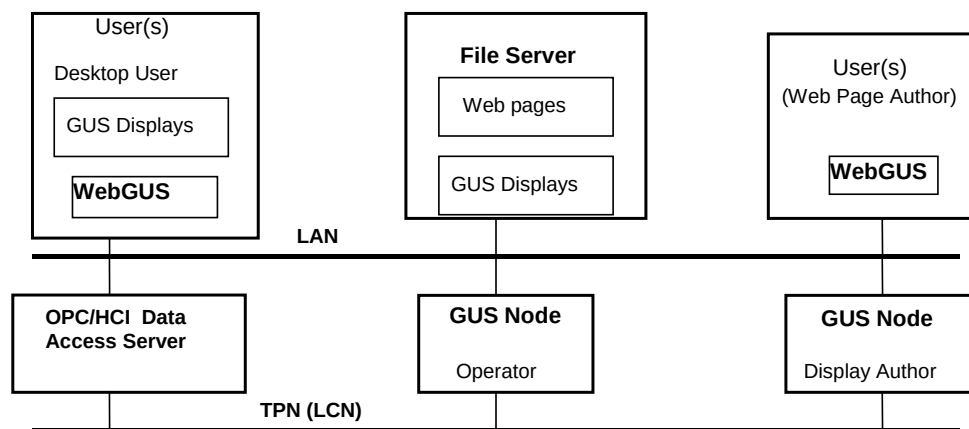
TIP

Often, the above issues may not be practically solved. As a workaround, an alternate solution based on the Windows 2000 or Windows 2003 Terminal Server may be used.

Example Data Redirection topology

In this configuration, WebGUS desktop users are able to view GUS displays which were originally created for TPN control room operations.

Figure 4 Data Redirection Topology



In this example, an OPC/HCI Data Access server collects LCN parameters and allows access to those parameters by WebGUS desktop users. This data server would usually be a HCI PHD server, but other types of servers are possible (for example, the HCI TPN Server).

GUS displays reside in a common repository on the network and can be read by WebGUS users.

GUS nodes usually access the displays locally, replicated from the common repository; in order to improve display call-up performance on GUS stations.

GUS displays are created on GUS (or GUS Remote Displays Client) nodes because full validation of displays requires connection to the LCN network.

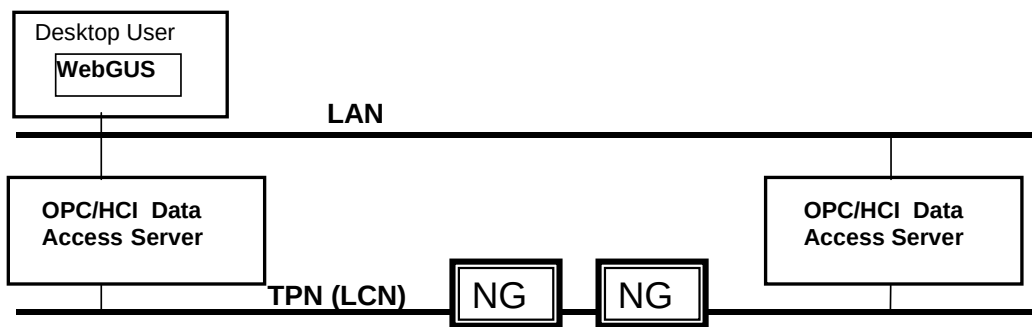
3 WebGUS Resource Topologies

3.1 Determining WebGUS Resource Topologies

Example multiple HCI/OPC server topology

In this example, each LCN network segment has its own surrogate HCI/OPC data server. LCN point names in GUS displays should be qualified with the LCN network gateway ID.

Figure 5 Multiple HCI/OPC Server Topology



4. Administering WebGUS

4.1 Overview of Installation and Configuration

Installation overview

A user in a WebGUS administrative role performs the tasks necessary to set up the system for desktop use. This role requires general knowledge of the following:

- GUS
 - As the human interface to an open control system.
 - As the integration of process operator and process engineer functions.
 - As the connection to the TPS plant information network.
 - As the use of standard and custom displays to view operating, alarm and system displays.
 - As the Display Builder tool for building and editing custom displays.
- LCN/HCI/OPC data server access as the ability to bind to an LCN/HCI/OPC server within a GUS display through the Named Data Access syntax to read and write the server data.
- Web page authoring and hosting
- Security and domain issues

4.2 Administering for Desktop Usage



ATTENTION

The WebGUS Administrative user consults with the local IT department, and Windows 2000 and networking experts when setting up the installation. The WebGUS Administrative user will either have necessary administrative permissions to fully install and configure the system or will delegate specific tasks to other authorized users.

Scope of administrative tasks

WebGUS administrative tasks include:

- Planning the installation by determining the topology and resources for the system.
 - In particular, the administrative user locates the site server computers, which will store WebGUS software distribution files, environment definition files, Web pages, and GUS displays.
- Installing WebGUS from the Honeywell software distribution CD on a site server.
- Preparing the WebGUS Download and Configuration Web pages.
- Creating common WebGUS configuration profiles.
 - Referred to as WebGUS environments.
- Defining the scope and topology for LCN Data Redirection.
 - LCN Data Redirection makes it possible for WebGUS users to view existing GUS displays. These displays access LCN data through the surrogate HCI/OPC data servers.
- Facilitating remote access between different WebGUS installations.
- Consulting with administrators of remote plants.
- Assigning user permissions for accessing WebGUS environments, GUS displays and Web pages.
- Defining a policy for releasing newly developed GUS displays, WebGUS Web pages, and Environment files into common areas.

4.3 Administering for HCI/OPC Data Access

A user in a HCI/OPC Data Access Server Setup administrative role performs the tasks necessary to set up the HCI/OPC Data Access servers for all WebGUS users. The servers are installed according to the instructions provided by the manufacturer of the server.



REFERENCE - INTERNAL

For more details, see the following documents:

- "Configuration Hierarchy" section in the *GUS HCI/OPC Data Access User's Guide*
 - "Customizing HCI/OPC Server Capability Files" section in the *TPS System Administration Guide*
-

HCI/OPC Data Access administrative tasks

HCI/OPC Data Access Server Setup administrative tasks include:

- Installing and configuring HCI/OPC data servers.
- Providing information to the WebGUS Administrator about the identity and location of the servers.
 - The administrative user needs this information to set up the environment configuration files.
- Resolving HCI/OPC security access issues.

LCN Data Redirection administrative tasks

If LCN Data Redirection is required, the HCI/OPC Data Access Server Setup administrative tasks include:

- Installing and configuring HCI/OPC data servers that provide TPN (LCN) tags to GUS displays through OPC.
- Defining the distribution of TPN segments into the surrogate HCI/OPC servers.
- Providing the mapping information for a TPN (LCN) server and its surrogate server to the WebGUS Administrator.
 - The administrative user needs this information to set up the environment configuration files.

4.4 Resolving Compatibility with Other Honeywell Software

Packages that can be installed

The Honeywell TPS System Software distribution CD contains other packages that may be used on the WebGUS computer.

The following packages may be installed with WebGUS:

- GUS Display Builder
- TPS domain OU infrastructure packages and HCI servers
- Other packages that have been declared to be compatible with WebGUS



TIP

Installation of the GUS OPC Client Add-In is not required, because WebGUS includes the HCI/OPC client connectivity components.

Installation and configuration of these packages usually requires administrative privileges, in particular open access to the HKEY_LOCAL_MACHINE registry hive.

Packages that are incompatible with WebGUS

The following packages on a WebGUS computer **ARE NOT SUPPORTED**:

- GUS Display Runtime
- GUS Multiple Displays
- GUS Remote Client



ATTENTION

Un-installation of any of the above packages may disrupt operation of WebGUS. WebGUS should also be removed and then reloaded through the WebGUS Download and Configuration page.

4.5 Install and Uninstall Rules for WebGUS and GUS Display Builder on a Desktop PC

Since the WebGUS ActiveX control is part of both the WebGUS package and the GUS Display Builder package, some rules apply when a Desktop PC has both WebGUS downloaded and the GUS Display Builder installed.

The download and installation order determines which WebGUS ActiveX is registered and how the Registry handles references to the locations for WebGUS's dependent DLLs. The order creates interdependencies that impact how the removal or uninstall cleans up the directories and registry references.

Failure to follow these rules could result in error messages displayed, such as

“The specified module could not be found”

while trying to use the products.

Rule 1

The prescribed order to follow to run both on a Desktop PC is;

- Download WebGUS
- Install the GUS Display Builder

Rule 1a

If the GUS Display Builder is already installed on the Desktop, remove the GUS Display Builder prior to downloading WebGUS and then re-install the GUS Display Builder.

Rule 2

If either WebGUS is removed or the GUS Display Builder uninstalled, the remaining WebGUS or GUS Display Builder package must be removed or uninstalled. For example, if you uninstall the GUS Display Builder, then you must also remove WebGUS and re-download WebGUS if you want to continue running WebGUS.

Rule 2a

The prescribed order to follow to remove both packages on a Desktop PC is:

- Uninstall the GUS Display Builder.
- Remove WebGUS

4.6 Authoring for WebGUS

GUS display authoring tasks

A user in the GUS Display Author role creates GUS displays using the GUS Display Builder. The GUS Display Builder may be installed on desktops and/or existing GUS nodes (including GUS Remote Displays clients).

A local policy, set by the WebGUS Administrator, regulates the release of the developed GUS displays into a common area accessible to desktop users.

Web page authoring tasks

A user in the Web Page Author role creates the Web pages that contain the WebGUS ActiveX control for presenting GUS displays in the Internet Explorer. The Web Page Author is proficient in creating Web pages, including page scripting and methods of interactions between various Web page components, and uses a desktop computer to test Web pages with embedded GUS displays.

The amount of communication between the Web Page Author and GUS Display Author depends on the amount of interaction between the Web pages and the GUS displays. Refer to the GUS Display Scripting User's Guide for GUS display scripting information.

A local policy, set by the WebGUS Administrator, regulates the release of the Web pages into a common area accessible to desktop users.

4.7 Configuring Desktops

Desktop user administrative tasks

A desktop user accesses the WebGUS Download and Configuration page to download and configure the WebGUS and the LCN Redirection Component to the desktop computer. A desktop user operates GUS displays within the Internet Explorer.

5. Applying Security to Support WebGUS Deployment

5.1 Interpreting WebGUS Security Context

A WebGUS user operates in the security context of the currently logged-on Windows 2000 user. Security settings that are configured for various WebGUS resources determine the user's rights to perform actions on these resources.

WebGUS Security requirement

The base requirement of WebGUS security is:



ATTENTION

Any server application that provides access to a protected resource (a GUS display file or an HCI Data Access server) must be able to positively identify the user that is requesting service.

Authentication scenarios

Verification of user identity (authentication) is performed by various system infrastructure components within Windows 2000 user domain structure.

In a basic scenario, all users and servers operate within a *common* Windows 2000 user domain and there are no authentication issues.

In more complex scenario, server applications may reside in different Windows 2000 domains. In order to successfully authenticate the identity of the user, the server's domain must trust the domain of the user.

5.2 Assigning WebGUS User and Group Security

Approach to user and group security

Security is best thought of in context of user roles:

- A particular user role has associated tasks.
- Tasks require sufficient security permissions configured for the user.
- Security permissions bound the limits of user actions by protecting controlled resources.

A user role may be mapped into Windows 2000 user-group membership. This simplifies management of security because a particular resource security privilege is assigned (or revoked) on a user-group basis, instead of a single-user basis. All individual users that belong to a user-group gain (or lose) particular permissions.

Providing a security framework

The HCI (Honeywell Communications Interface) infrastructure and TPS security model provide a security framework that can be used in WebGUS. Alternatively, the WebGUS administrator can devise user-group membership schemes that will form the basis for managing security in WebGUS.



REFERENCE - INTERNAL

See "Applying TPS Security Model" in the *TPS System Administration Guide*.

5.3 WebGUS Security

The Windows operating system and the configuration pages of the OPC/HCI Data Access servers provide tools for applying security permissions.

Server host computers

Server host computers encompass:

- Computers that store common WebGUS resource files.
- Computers that host OPC/HCI Data Access servers.

The computer that hosts a server must allow particular user-groups of WebGUS users to access the computer remotely. This permission is assigned using the Windows 2000 Group Policy snap-in.

WebGUS user computers

A WebGUS user must be able to download and configure WebGUS to a desktop computer. The Internet Explorer downloads and configures all WebGUS executable components locally. The WebGUS control is signed with the Honeywell trust certificate and is declared safe for initialization and scripting, thus removing the need to operate Internet Explorer at lowered security settings.

The WebGUS Administrator, in conjunction with the local IT department, must ascertain that the IT policy for desktop users allows downloading and registration of ActiveX controls under Internet Explorer.

The account used by the HCI/OPC Data Access server to do data-change-event callbacks must have the user right, "access this desktop computer from network", on the desktop computer. The Group Policy snap-in sets this user right.



ATTENTION

This action may not be needed if the server can operate in a one-way domain trust context.

An administrative user, with proper security permissions, will perform the initial download of the WebGUS to the desktop (for example, to gain access to the HKEY_CLASSES_ROOT registry hive). The original desktop user will then configure WebGUS for use by selecting the proper WebGUS environment.

5 Applying Security to Support WebGUS Deployment

5.3 WebGUS Security

The download and configuration of WebGUS components modifies hives in the following local computer registries:

- HKEY_CURRENT_USER -- stores WebGUS environment configuration data.
- HKEY_LOCAL_MACHINE



ATTENTION

The user will need Read/Write access to the WINNT\System32 directory to perform the download WebGUS.

HCI/OPC Data Access servers

The OPC/HCI Data Access servers are COM (formerly known as OLE) objects. The security mechanisms of COM apply in OPC. In order to use an OPC server, the user must be assigned access (and launch, if required) permissions at the server computer. The Windows 2000 DCOMCNFG tool's security page can be used.

DCS system

Access level to the DCS is the responsibility of the server. HCI servers can be configured for View Only.

If a particular non-HCI OPC Data Access server does not provide the View-only capability, then WebGUS scripting may be used to force the View-only operation. Such a solution is less secure because it is implemented on the client side.

5.4 WebGUS File Security

Common resources files

The common resource file category includes:

- GUS displays
- WebGUS environment files
- WebGUS Web pages
- WebGUS distribution files

Providing a security framework

Qualified user-groups may be assigned the right to read particular files or folders. Without the Read permission, the file is invisible to the user; programmatic attempts to read the file will result in errors.

A very limited group of users may be assigned the right to Write/Modify. Only these users will be able to update contents of the common resource. The Read, Write and Modify permissions can be assigned using the Security page of the Windows 2000 File Properties page.

Usually stored on file servers, Web pages may be under control of the Web server. The Internet Information Server (IIS) is a service that runs on top of Windows 2000. It relies heavily on user accounts and the NTFS file system.

When IIS is installed it creates two user accounts, assigns them specific rights, and places them in user groups. One account, IUSR_computername is used to grant anonymous access to Web resources. Every anonymous visitor to the Web site will be given the rights granted to the IUSR_computername account. For sites with confidential information such that anonymous access is deemed inappropriate, IIS offers other authentication options.

IIS has Web permissions such as Read and Write. The NTFS file system has its own Read and Write permissions. NTFS permissions control user accounts and the type of access to resources on the hard disk. Web permissions control the HTTP verbs and their execution on the HTTP resources. HTTP verbs are the commands sent in the headers from browsers requesting actions on resources. For example, they might say: "I want to read this Web page".

5 Applying Security to Support WebGUS Deployment

5.4 WebGUS File Security

The Read and Write check boxes on the Web Site tab in IIS control which HTTP verbs are allowed for the resources. They do not control the NTFS file system permissions for the resources. If Web and NTFS permissions do not agree, the more restrictive of the two will be used for HTTP requests. Therefore, access to Web pages and graphics can be limited by setting directory and file permissions on the .htm, .asp, or .gif files.



TIP

You will see better performance by replicating the GUS display to a local drive.

6. Configuring the WebGUS Environment

6.1 How WebGUS Environments Will Function

Creating WebGUS environment definitions

WebGUS environments may be created and saved to a persistent file using the WebGUS Download and Configuration interface supplied with the product. The purpose of this interface is to:

- Download the WebGUS control to the desktop.
- Manipulate the WebGUS desktop operating environments.

The interface consists of two Web pages. The pages are identical in appearance but differ in purpose and targeted user:

- The first, the WebGUS Download and Configuration page, can both download WebGUS and configure the environment. The targeted user of the dual-purpose page is an administrator.
- The second, the WebGUS Environment Configuration page, can only configure the environment. The targeted user of the single-purpose page is a desktop user.

Each page has a different capability based on the HTML OBJECT tag syntax used on the page. The OBJECT tag identifies the WebGUS control. The OBJECT tag attribute, CODEBASE, specifies the location on the Web server for the file containing its source, packaged for download:

- The WebGUS Download and Configuration page includes the CODEBASE attribute.
- The WebGUS Environment Configuration page does not.

6 Configuring the WebGUS Environment

6.1 How WebGUS Environments Will Function

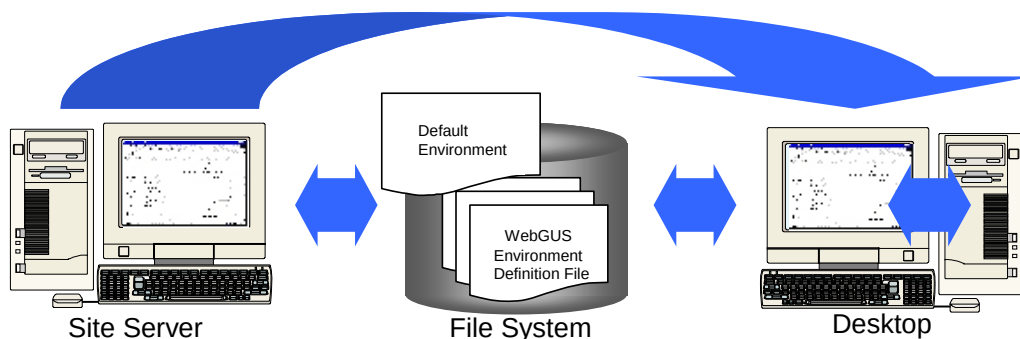
Constructing default environments

When a customer installs WebGUS on a site server, a WebGUS Administrator or other knowledgeable user will construct one or more WebGUS environment definitions and save them in persistent files, accessible to the desktop users that need them. The administrator will designate one of the environment definitions as the default environment. Part of the setup for the WebGUS Download and Configuration page at the site server is specifying the URL of the default environment.

Applying the default environment

The initial access of the WebGUS Download and Configuration page by the desktop user will download the control, access the default environment, and apply the default environment to the registry settings for the currently logged-on account. After the page has applied the settings to the registry, the default environment becomes the current environment.

Figure 6 Download of Default Environment Definition



ATTENTION

WebGUS Environments may also be defined by entering an XML file. The format for the XML is file is published as part of the WebGUS product and can also be found in this document under Consolidated DTD.

Browsing and viewing environments

The WebGUS Download and Configuration page contains a Browse for Environment button providing access to a standard Open File Dialog, which allows any user to browse accessible files to locate the current environment files or other environment definitions.

This will display the settings relevant to the appropriate component in the Environment Settings portion of the page.

For the WebGUS component, the settings include a table of the OPC Data Access servers, and a list of GUS runtime parameters. For the LCN Redirection Component, the settings include a table of server maps and renaming conventions.

Changing the current environment

The WebGUS Download and Configuration page allows the desktop user to browse to any existing environment definition and make it the current environment. The desktop user selects the file with the environment definition and clicks the Set As Current Environment button. The page reads the settings from the environment definition and applies them to the registry at the desktop.

Modifying an environment definition

The WebGUS Download and Configuration page allows the user to modify the settings for any existing accessible environment definition and save the modifications to a file as a new environment definition.

The page allows addition, deletion and modification of fields on the page. The user may click the Save As button, which provides the standard Save As dialog to save the environment definition in a location accessible to the user.

Displaying current environment

The WebGUS Download and Configuration page allows the user to view the settings for the current environment. The Show Current Environment button applies the settings in the desktop registry keys to the environment settings on the page.

6.2 Using the Environment Configuration Interface

The WebGUS Configuration Web page bundles all of the runtime settings necessary for execution of GUS displays that are embedded in Web pages. The bundle of runtime settings constitutes an Environment Definition. The page stores current environment data within the scope of the current Windows user.

WebGUS environment settings precede other sources of configuration settings, such as configuration of HCI Data Access servers.

A desktop user accesses the WebGUS Configuration page to download and configure WebGUS and the LCN Redirection Component on the desktop computer.

Figure 7 Environment Configuration Interface Web Page

Name	Prog ID	Host Computer	Description	Collection Rate(sec)	Ping Parameter
AMWB2_TPNServer	Hci.TPNServer	xyz-aa2-S009DE	TPN Server for Unit 1	10	
AMWTB2_PHDServer	Hci.PHD	system57	Plant History Data for Unit	60	
AMPLC_OPCTServer	{ADF6AE8B-B0F1-11d0-8000-000000000000}	hw-xyz-bcm010	an OPC DA server		

6 Configuring the WebGUS Environment

6.2 Using the Environment Configuration Interface

LCN Data Redirection

Rename Point

Insert Prefix:

Redirect Point

Point Prefix	HCI / OPC Surrogate Server
G1*	AMWB2_TPNServer
G2*	AMWVB2_PHDServer
*	AMWB2_TPNServer

Runtime Timeout Settings:

(Note: Changes to timeout settings may impact the way the system operates and should only be made by an expert user.)

HCI/OPC Access:

Bind Time(msec):

Ping Time(msec):

Access Time(msec):

Script Processing:

UI Execute(msec):

Data/Periodic(msec):

Cursor Change Response(msec):

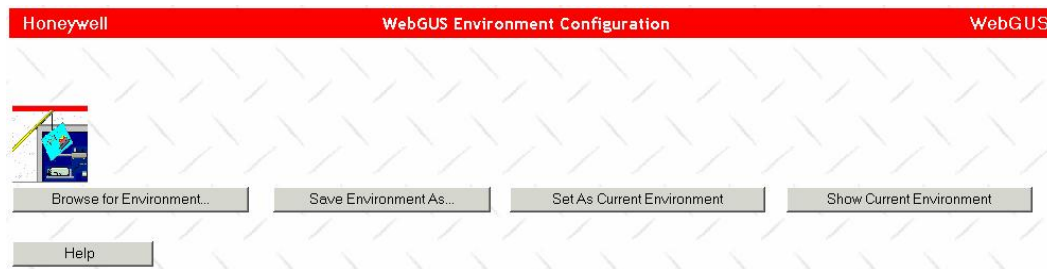
Shutdown Wait Limit(msec):

Allow Unvalidated Displays:

6.3 Using the Environment Selection Section

The Download and Configure Section page downloads the WebGUS ActiveX control and the LCN Redirection Component and provides the interface to manipulate an environment definition.

Figure 8 WebGUS Configuration Page HMI – Download & Configuration



The page contains the following buttons:

Button	Function
Browse for Environment	Provides the standard File Open dialog for the user to browse any accessible portion of the file system and select any accessible Environment Definition.
Save Environment As	Provides the standard Save As dialog for the user to save Environment Definitions.
Set As Current Environment	Applies the Environment Settings on the page to the desktop registry.
Show Current Environment	Displays the settings in the desktop registry keys on this page.
Help	The Help button invokes the standard Microsoft HTML Help, dual-pane interface. The Help interface briefly describes the WebGUS Environment Configuration page and its contents. The Help also has links to more in-depth information. These links will only work if the user has installed the Honeywell TPS Knowledge Builder on their PC.

Whether a user has the capability of performing any of these functions directly relates to user security permissions. Refer to Interpreting WebGUS Security Context for an extensive discussion of user roles and security.

Setting the default environment as the current environment

If you want to restore the default environment settings that were installed when you launched the WebGUS page for the first time, follow the steps in the table below.

Step	Action
1	Click the Browse for Environment button.
2	Browse to the location that your WebGUS Administrator has established as the default Environment location.
3	Select the appropriate Environment.
4	Click Set As Current Environment button.

6.4 Using the Environment Settings Section

Assigning a Display Search Path

Figure 9 WebGUS Configuration Page HMI – Environment Settings



The Display Search Path is a semi-colon (;) separated list of folders that contains GUS displays. Syntactically, the list of folders should not contain blank characters following the ‘;’ character. The Display Search Path has a maximum length of 255 characters.



ATTENTION

If the user makes a request to run a display without the full path name for the display file, the GUS display runtime engine searches the list of Display Search Path folders in the specified order.

Display search path example

If the parameter is:

Display Search Path = C:\TestDisplays;\\xyz-srvr1\GUSDisplays\Area5

and the user invokes the display with

WebGUS.display = “TOWER50”

Then the GUS Display Runtime Engine searches the “C:\TestDisplays” folder first, then \\xyz-srvr1\GUSDisplays\Area5 for the file: *TOWER.PCT*.

If the file *TOWER.PCT* exists in folder C:\TestDisplays, then the search stops.

Entering and editing a Search Path

To enter a new search path or edit an existing search path, follow the steps in the table below.



ATTENTION

Make sure you use a semi-colon (;) between search path locations and insure that you type the path exactly the way it appears in the directory structure, including spaces and backslashes where appropriate to delineate directories.

Step	Action
1	In the Display Search Path box, highlight the path you want to edit or, click the end of the existing search path string to add a new search location. If you are adding a new search location, ensure you include a semi-colon at the end of the existing search path string.
2	Make your changes and/or additions.
3	Click the Set As Current Environment button. Note: See the section "Using the Environment Selection Section" in this guide for additional information.

6.5 HCI/OPC Data Access Servers

Installation of the OPC Data Access server is independent of WebGUS and proceeds according to instructions from the manufacturer of the server. The HCI data access servers may be installed in the TPS domain OU.

Installation of an OPC/HCI server usually consists of two steps:

1. Installation at the server host computer's server side.
2. Installation at the computer's client side.

The client-side installation of the server is not required on WebGUS computers.

WebGUS includes the standard OPC Foundation proxy files, as well as the standard HCI proxy files (generated by Honeywell). These proxies are sufficient for WebGUS to access any OPC/HCI Data Access server (up to Version 2.04).



REFERENCE - INTERNAL

For more detail on implementation of the WebGUS HCI/OPC client, refer to the *GUS HCI/OPC Data Access User's Guide*.

WebGUS requires HCI/OPC server PROGID or CLASSID

Definition of an OPC/HCI server in the WebGUS environment requires providing the CLSID or PROGID of the server.

- CLSID is the actual server object identifier
- PROGID is a user-friendly name, usually needed by the VB applications (for example, CreateObject("progid")). Internally, PROGID must be converted into CLSID before accessing the server.

Well-known PROGIDs of selected HCI Data Access servers are built into WebGUS. For these IDs, WebGUS software can resolve CLSID internally. Also, any server's CLSID may be provided directly. Refer to the description of the WebGUS Download and Configuration page.

Installation of the client-side server usually registers the server's PROGID and may be required for the benefit of other applications, possibly to include other non-standard OPC extension proxies.

The WebGUS environment definition accepts PROGIDs that are provided by these installations.

Assigning HCI/OPC Data Access servers

Figure 10 HCI/OPC Data Access Servers Configuration

HCI / OPC Data Access Servers					
Defines, supplements, or overrides server definition by TPS domain.					
Name	Prog ID	Host Computer	Description	Collection Rate(sec)	Ping Parameter
AMWB2_TPNServer	Hci.TPNServer	xyz-aa2-S009DE	TPN Server for Unit 1	10	
AMWB2_PHDServer	Hci.PHD	system57	Plant History Data for Unit	60	
AMPLC_OPCTServer	{ADF6AEBB-B0F1-11d0-t	wi-xyz-bcm010	an OPC DA server		

The middle section of the WebGUS Environment Configuration page is the Environment Settings where HCI/OPC Data Access servers are selected.



ATTENTION

If the WebGUS computer is a member of the TPS Domain OU, then the managed HCI components within that OU are accessible from WebGUS, independent of this list.

This configuration can be used to:

- Define additional servers.
- Override **PROGID** and **HostComputer** for the server name (in effect redirecting data access to another server).
- Override the data collection rate on a connection to the server.

Purpose of server table

The HCI/OPC Data Access server table defines an association between a server name, a server object ID and a server location. The table contains five fields that are described below.

Field	Description
Name	GUS display scripting recognizes an HCI/OPC data server by its name. A GUS display author provides this name explicitly in the GUS script HCI "BIND" statement. Hci.server.bind "Name" REFERENCE: See the "Binding and Unbinding HCI/OPC Servers" in the <i>GUS HCI/OPC Data Access User's Guide</i> for details. When defined through TPS configuration tools, the name is referred to as the HCI Alias Name. If a particular server name cannot be resolved, then GUS displays that access this server will present communication errors.
Prog ID	The PROGID is the COM object ID of the server; presented as a PROGID or CLSID. This uniquely identifies the type of HCI/OPC Data Access server. The GUS display runtime engine automatically recognizes well-known PROGIDs of the selected HCI/OPC servers, such as: "Hci.TPNServer" "Hci.PHD" Instead of the PROGID, the user can enter the server's CLSID directly. The format of the CLSID must include the enclosing braces: "{.....}".
Host Computer	The HostComputer is the network location of the computer that is hosting the server.
Description	The Description is explanatory text that describes the server's functionality/location/manufacture and so forth. The GUS display runtime engine does not use this field. This is an optional field.

Field	Description
Collection Rate (sec)	The CollectionRate is the override of data collection rates built into GUS displays. If the field remains empty, then there is no override and the collection rates are unchanged. Collection rate overrides are entered in seconds. The value of 0 is equivalent to “collect once – only”, during GUS display startup. The collection rate override applies only on the connection between a particular WebGUS and a server. It does not affect other clients of the server.
PingParameter (optional)	The PingParameter is a string data parameter that is provided by the server, periodically reporting change of value. If the parameter value does not change at least once during the “Ping” time (a WebGUS runtime parameter), then the server will be considered off-line. The “PingParameter” mechanism provides an improved real-time diagnostic of the data change path between an OPC/HCI data server and a WebGUS display. Note: The WebGUS OPC/HCI communication infrastructure provides other, simpler mechanisms between client and server, independent of the “PingParameter” option. Internally, WebGUS will request that the “Ping Parameter” be collected at a 4-second update rate.



TIP

The pair **ProgId/HostComputer** uniquely identifies a particular server.

Examples of HCI/OPC Data Access Server Binding

Example 1 Binding to HCI/OPC Servers

The scripting in GUS displays binds to any of the servers: Server1, Server2 and Server3. Server1 and Server2 are HCI TPN Servers. Server3 is an OPC DA server provided by a third-party manufacturer. The PROGID for Server3 has not been installed, but CLSID is known. GUS displays should collect data at rates specified by the displays. A TPS Domain OU is not present.

The GUS scripts are:

- Hci.s1.Bind "Server1"
- Hci.s2.Bind "Server2"
- Hci.s3.Bind "Server3"

The resulting configuration is:

Name	PROGID	HostComputer	Description	CollectionRate (sec)
Server1	Hci.TPNServer	xyz-abc-123		
Server2	Hci.TPNServer	xyz-abc-456		
Server3	{...clsid string...}	xyz-abc-789		

Example 2 – Binding to TPS Domain OU servers

The WebGUS computer is in a TPS Domain OU. Scripting in GUS displays binds to the servers that are managed by the TPS Domain OU. GUS displays should collect data at rates specified by the displays.

The resulting configuration is:

Name	ProgID	HostComputer	Description	CollectionRate (sec)

Example 3 - Overriding the TPS Domain OU server definition

- **Example 3a**

The WebGUS computer is in a TPS Domain OU. Scripting in GUS displays binds to the servers that are managed by the TPS Domain OU.

In this example, a *TPNServer1* server is defined as an Hci.TPNServer located at tpnsrvr_computer.

This WebGUS user should not access TPN data directly through this server because of data load and security access considerations. Instead, the user may reference a surrogate server of the Hci.PHD type, located on a phdsrvr_computer, which has TPN tags mapped. Additionally, this WebGUS client will collect data at a 30-second rate from this server, regardless of collection rates programmed into displays.

Definition of other servers (if present) remains unchanged.

The GUS script(s) are:

- Hci.s1.Bind “TPNServer1”

The resulting configuration is:

Name	ProgID	HostComputer	Description	CollectionRate (sec)
TPNServer1	Hci.PHD	phdsrvr_computer		30

Example 3b

This is a variation on the previous example. Here, the WebGUS user accesses the TPNServer1 as configured by the TPS Domain OU. This display will be a one-time snapshot. (The data is collected only at display startup).

The GUS script is:

- Hci.s1.Bind “TPNServer1”

The resulting configuration is:

Name	ProgID	HostComputer	Description	CollectionRate (sec)
TPNServer1	Hci.TPNServer	tpnsrvr_computer	...	0

6.6 Setting Environment for LCN Data Redirection

Purpose

The LCN (TPN) Data Redirection section of the configuration identifies HCI/OPC servers, which act as surrogate providers for TPN name forms. Additionally, this section of the configuration defines the optional prefix to be inserted in front of the TPN name form.

Figure 11 LCN Data Redirection Configuration



Point Prefix	HCI / OPC Surrogate Server
G1*	AMWB2_TPNServer
G2*	AMWTB2_PHDServer
*	AMWB2_TPNServer



ATTENTION

- GUS display scripts are not required to explicitly bind to the surrogate servers for the purpose of LCN Data Redirection. Bind, in this case, is automatic.
- WebGUS does not support dynamic redirection.
 - If you change the HCI/OPC server name and LCN mapping entry on the WebGUS configuration page, the change will not take affect until you re-load the web page that contains the GUS display WebGUS server reference.
 - Therefore, if you use the WebGUS Configuration page to make a change to your current environment that will affect a running GUS Display, the change will not take affect until you close the display and reload it.
- If surrogate server mapping information is not provided, GUS displays will attempt to access LCN data through regular GUS HOPC Data Access channels. This results in runtime errors in an environment that does not have access to the LCNP board. The existing GUS scripting handles the runtime errors.

Rename point

This optional field defines the prefix to be inserted in front of the LCN point tag name. If the surrogate server mapping information is not defined, then the prefix is disregarded.

On a given LCN tag name, WebGUS Data Redirection Component inserts the prefix directly after the “.” that follows the “LCN” string.

For example:

LCN.Name > LCN.*PrefixName*

The prefix is not inserted if it ends with the “\” character and the given LCN point tag name already has the “\” character in it.

The WebGUS LCN Data Redirection Component inserts the prefix first, and then attempts to find the surrogate server matching on the Point Prefix name.

Re-direct point

This section of the Configuration page assigns the surrogate server based on the TPN (LCN) point tag name form.

Field	Description
Point Prefix	The Point Prefix field is the GUS scripting LCN name form, without the “LCN.” prefix. The name form may include an asterisk (*) or consist only of an asterisk as a wildcard.
HCI/OPC Surrogate Name	The HCI/OPC Surrogate Name is the name of the HCI/OPC Data Access server that provides data for LCN points that begin with the Point Prefix name. The server name must be configured for use at this WebGUS computer (either through the HCI/OPC Data Access server section of the Configuration page, or by the TPS Domain OU configuration).



TIP

The prefix would usually indicate LCN network gateway ID; the backslash “\” character must then be included.

6 Configuring the WebGUS Environment

6.6 Setting Environment for LCN Data Redirection

On a given LCN point tag name, the WebGUS Data Redirection Component attempts to match the first characters of the name with the defined Point Prefix strings. The rules used for matching are:

- The matching is case-insensitive.
- If prefixes are present, they will be tested first.
 - Testing starts with the longest prefix in order to obtain the most specific match.
- If provided, the standalone entry “*” will be tested last.

If the tag name cannot be matched then an empty server name is returned to WebGUS display runtime engine. No log is produced of un-matched tag names.

If HCI/OPC server is unknown then the GUS display runtime scripts produce communication errors on access to the LCN name.

If an error occurs when attempting to bind to a surrogate HCI/OPC server, an error is displayed, which is the same error displayed when this occurs for standard GUS displays. Similarly, if the bind is successful, but the surrogate server does not support the requested data point, an error is displayed which is the same error when this occurs for standard GUS displays.

Example of mapping to PHD server

A PHD server should service all LCN name forms present in GUS displays that execute in WebGUS. WebGUS presents all names to the PHD server unchanged.



ATTENTION

The PHD is shown for example purposes only. Other server types may also act as surrogate providers of the LCN tags.

The configuration is:

HCI/OPC Data Access Servers

Name	ProgID	HostComputer	Description	CollectionRate (sec)
PHD	Hci.PHD	phdsrvr_computer	LCN Surrogate	30

Surrogate Server Mappings

Point Prefix	HCI/OPC Surrogate Name
*	PHD



TIP

The Uniformance PHD Server API provides an additional tag name translation function for connected clients called "Tag Name Aliases". This function allows prefix and suffix filtering, as well as the capability of renaming specific names. Refer to the documentation for the PHD server.



ATTENTION

The CollectionRate override on a connection to the PHD server is not necessary, but most likely needed to avoid execution of GUS displays at rates programmed for control.

Example of mapping segments to PHD

Two segments of the LCN map to separate PHD Servers, PHD1 and PHD2. The corresponding LCN network gateway IDs are: N1 and N2. Most of the LCN name forms present in GUS displays are well qualified by the network gateway ID; however, early GUS displays do not include the LCN network gateway ID. These displays reference points from segment N1.

The configuration is:

HCI/OPC Data Access Servers

Name	ProgID	HostComputer	Description	CollectionRate (sec)
PHD1	Hci.PHD	phdsrvr_computer1	LCN Surrogate N1	30
PHD2	Hci.PHD	phdsrvr_computer2	LCN Surrogate N2	30

Surrogate Server Mappings

Point Prefix	HCI/OPC Surrogate Name
*	PHD1
N1*	PHD1
N2*	PHD2

Or:

Point Prefix	HCI/OPC Surrogate Name
N1*	PHD1
N2*	PHD2

And:

“InsertPrefix” = “N1\”



ATTENTION

The way tags have been configured in the PHD Server may dictate whether or not you choose to insert a prefix. In the first case, a “*tagname*” is requested from the server; in the second case, the “N1*tagname*” is requested.

Example mapping requiring custom component

Two segments of the LCN network have been mapped to separate PHD servers: PHD1 and PHD2. The LCN network segment gateway network IDs are: N1 and N2. Most of the LCN name forms present in GUS displays are well qualified by the network gateway ID. However, there are early GUS displays on which the LCN names do not include the LCN network gateway ID. These displays are known to reference points either from segment N1 or N2.



TIP

The key problem is that there is no clear way of saying which server is supposed to map for the unqualified LCN names; it can be N1, or N2.

The standard version of the LCN Data Redirection Component **does not solve** this problem. Possible solutions are:

- Update GUS displays to fully qualify LCN point names with LCN network gateway IDs.
- Replace the standard LCN Data Redirection Component with a customized version. The new version would insert the network gateway ID based on location (full path file name) of the GUS display. Displays that access N1 would need to be grouped into a different file folder from those displays that access N2. The location of the display is provided to the Data Redirection Component with each LCN point name.

The LCN Data Redirection Component is a Visual Basic application that matches LCN point names to surrogate servers, based on parameters stored in the local registry. The GUS display runtime engine accesses this server application through a single (dispatch interface) method. Source code for the project is distributed with WebGUS on a Honeywell Software Distribution CD.

The configuration parameters of the new, customized LCN Data Redirection Component may be different from those present on the standard WebGUS Download and Configuration page. In this case, the LCN Data Redirection section must be updated for each requirement of the new LCN Data Redirection Component. The new customized component may define its own way of storing the redirection parameters. The only “constant” is the method name and parameters of the LCN name-matching function that the component provides to WebGUS.

6.7 Modifying an Environment Setting

The Runtime Timeout Settings are tune-up timing parameters for the GUS display runtime engine. If no specific reason exists to specify values for these parameters, the entries should be left unchanged to allow the use of internally programmed default values.

Figure 12 WebGUS Configuration Page HMI – Modifying Runtime Settings

The screenshot displays the 'Runtime Timeout Settings' section of the WebGUS configuration interface. It features a green header bar with the title 'Runtime Timeout Settings:'. Below the header, a note states: '(Note: Changes to timeout settings may impact the way the system operates and should only be made by an expert user.)'. The settings are organized into two columns. The left column, titled 'HCI/OPC Access:', contains three input fields: 'Bind Time(msec):' with a value of 20000, 'Ping Time(msec):' with a value of 30000, and 'Access Time(msec):' with a value of 10000. The right column, titled 'Script Processing:', contains four input fields: 'UI Execute(msec):' with a value of 0, 'Data/Periodic(msec):' with a value of 0, 'Cursor Change Response(msec):' with a value of 500, and 'Shutdown Wait Limit(msec):' with a value of 10000. At the bottom right, there is a checkbox labeled 'Allow Unvalidated Displays:' which is currently unchecked.

Setting Category	Parameter	Value
HCI/OPC Access:	Bind Time(msec):	20000
	Ping Time(msec):	30000
	Access Time(msec):	10000
Script Processing:	UI Execute(msec):	0
	Data/Periodic(msec):	0
	Cursor Change Response(msec):	500
	Shutdown Wait Limit(msec):	10000
	Allow Unvalidated Displays:	☐

All values in the following table are entered in milliseconds.

HCI/OPC Access

Setting	Description
Bind	The Bind timeout indicates the amount of time spent on waiting for a successful connection to an HCI/OPC server (execution of the scripting Bind method). If the server does not bind within Bind time, server failure is assumed. - Minimal value: 5000 msec. - Maximum value: 600000 msec. - Default: 20000 msec.
Ping	The Ping interval indicates the interval of polling a connected HCI/OPC server for a health status. A poll is executed using a read method that is governed by the Access timeout. Each server is being polled independently. If a response to a poll request is not received within the Access time, server failure is assumed. - Minimal value: 5000 msec. - Maximum value: 600000 msec. - Default: 10000 msec.
Access timeout	The Access timeout indicates the maximum amount of time spent waiting for a response from any method executed on a connected HCI/OPC server. Those include parameter read and write operations, among others. If a method does not return within the Access time, server failure is assumed. - Minimal value: = 5000 msec - Maximum value: = 600000 msec - Default: = 10000 msec

Script Processing

Setting	Description
UIExecute	The UI Execute timeout is the maximum time to allow a single GUS Display user interface script to execute before presenting a standard runtime error dialog message: "UI script execution failed to complete within configured time." On timeout, the GUS Display presents a standard script runtime error message box, with an option to close the display. • 0 = INFINITE timeout (hung script hangs execution of displays within the WebGUS control). (Default = 0 msec). Units are in msec.
Data/Periodic	The Data/Periodic timeout is the maximum time to allow a single DataChange/PeriodicUpdate script to execute before presenting a runtime error message: "The script execution failed to complete within configured time." On timeout, the GUS Display presents a standard script runtime error message box, with an option to close the display. Units are in msec.
Cursor Change Response	The Cursor Change Response timeout is the rate at which the target cursor change checks are made. Reducing this value improves cursor response at the cost of CPU-performance during cursor movement. • 0 = DISABLED (no cursor change over target). (Default = 500 msec.)

Setting	Description
Shutdown Wait Limit	The Shutdown Wait Limit timeout is the maximum time to allow a GUS Display to close normally before forcing the display process to terminate. If the display does not close within the Shutdown Wait Limit time, the GUS display runtime engine is terminated and the WebGUS control presents the error. To recover, the container display (or Web page) must be re-invoked. • 0 = INFINITE timeout (never force termination). (Default = 10000 msec.)
AllowUnvalidatedDisplays	The AllowUnvalidatedDisplays parameter is a Boolean flag, which indicates whether the user can run an un-validated display. A non-zero value will enable the execution of not fully validated GUS displays. An example is a GUS display that is created on a desktop and has LCN parameters. Validation of such display under GUS Display Builder on the desktop will produce LCN parameter validation errors. However, in most of the cases (where additional internal parameter definition is not required), for test purposes, this display may execute on a desktop accessing LCN parameters through an LCN surrogate HCI/OPC server. Note: A display must always be validated under GUS Display Builder (even though it may result in parameter validation errors), because validation of the display also compiles display scripts and creates the display symbol table.

6 Configuring the WebGUS Environment

6.7 Modifying an Environment Setting

7. WebGUS Environment XML Data Files

7.1 XML Detailed Description

Intended audience

The intended audience for this section is anyone interested in customizing the Re-direction Component.



ATTENTION

You do not need to know the following information if you are using the standard 'out-of-the-box' Honeywell implementation.

Overview

XML is a standard way to delimit text data. XML uses tags to convey the structure of the data. The tags, which delimit XML content, also name the specific element of data they delimit. Therefore, data is self-describing, since each item carries its own name and relates to the external model of a real world problem.

WebGUS uses XML to create a vocabulary for structuring information for the environment definitions. The WebGUS Download and Configuration page reads and writes environment data using XML files. The format of each section is clearly understood and corresponds naturally with the page data display and the System Registry settings.

The syntactic rules for the environment vocabulary are specified in a Document Type Definition (DTD). Since it is possible to create an environment definition without using the WebGUS Download and Configuration page, the DTD provides a mechanism for the page to do syntax checking for any definition.

7 WebGUS Environment XML Data Files

7.1 XML Detailed Description

Registry and XML conventions

The following conventions are used with regards to the WebGUS use of XML and the System Registry.

Data Exchange Format

The medium of exchange between the WebGUS control and the WebGUS Configuration Web page is a String. The String represents either:

- The XML file name
- The string content of an XML file stream

The XML file stream content string always relies on the Environment DTD for parsing correctness. Each Environment Definition has a processing instruction defining its DTD. The existence of the processing instruction guarantees that parsing occurs automatically by using the XML COM interface *load* and *loadXML* functions. WebGUS and its Configuration Web page need only provide formatting code for any parsing errors that occur.

Alphabetic Ordering Convention

The portion of the Environment Definition, relating to the Redirection Component, is customizable. In order for the registry to XML/XML to registry translation functions to be effective in the context of an unknown registry structure and an unknown Environment DTD, the functions need an alphabetical convention for ordering keys and document elements.



TIP

The intent of the **bold** and *italics* convention for **ELEMENT** and *attributes* in the following explanation and the subsections associating the registry, the Document Type Definition and the Environment Definition, is to establish a correspondence between the explanation and the relevant components.

The XML Environment Document Type Definition defines a positional location for each document **ELEMENT**. To parse correctly, the **ELEMENTs** in an Environment Definition must conform to the positions specified in its Document Type Definition. The XML *attributes* (**ATTLIST**) for an **ELEMENT** can appear in any order in the Environment Definition.

WebGUS functions processing Registry **keys**, which translate to XML document **ELEMENTs**, insert the **ELEMENTs** into the document in alphabetical order. WebGUS functions processing Registry **key values**, XML document **ELEMENT attributes** (**ATTLIST**), insert them in any order.

Naming Convention

The naming convention between the Registry keys and the XML document tags is, in most cases, a one-to-one correspondence. The name for a Registry key is the name of the XML **ELEMENT** tag. Similarly for the key values, the name of the value is the name of the *attribute* (ATTLIST). The only exception is the collection of OPCDA Servers and the LCN Mappings. To illustrate, the XML format allows for multiple tags with the <OPCDAServer> name while the Registry representation, OPCDAServer1, OPCDAServer2, and so forth, requires a numeric suffix to distinguish multiple keys with the same format.

To convert between the two representations when a collection exists, the DTD defines a *Count attribute* (ATTLIST) on the document **ELEMENT** bounding the elements of the collection. The WebGUS functions use the *Count* attribute and the bounding element to add/remove a numeric suffix.

XML Translation Key

The convention for processing the registry tree follows:

- A Registry key is an XML ELEMENT
- A Registry key value is an XML attribute (ATTLIST)

Example of Data Access Redirection in XML

XML Environment DTD

Below is the Document Type Definition for the Data Access Redirection XML vocabulary.

```
<!ELEMENT Environment (DARedirection, OPCDAServers, RuntimeParameters)>
<!-- Attribute for Environment -->
<!ATTLIST Environment FileSource CDATA #REQUIRED>
<!-- DARedirection Section -->
<!ELEMENT DARedirection (Honeywell)>
  <!ELEMENT Honeywell (LCNMappings)>
    <!ELEMENT LCNMappings ( LCNMapping* )>
    <!ATTLIST LCNMappings          Count    CDATA #REQUIRED
                                PointPrefix CDATA #IMPLIED >
    <!ELEMENT LCNMapping EMPTY >
    <!ATTLIST LCNMapping Prefix    CDATA #IMPLIED >
                                SurrogateServer CDATA #REQUIRED>
```

Where:

An Environment **ELEMENT** has three sub-**ELEMENT**s in alphabetical order:

- DARedirection
- OPCDAServers
- RuntimeParameters

The **Environment ELEMENT** has one *attribute* named *FileSource*.

The **DARedirection ELEMENT** has one **ELEMENT: Honeywell**.

The **Honeywell ELEMENT** has one **ELEMENT: LCNMappings**.

The **LCNMappings ELEMENT** has multiple **ELEMENT**s, each one an **LCNMapping ELEMENT**.

The **LCNMappings ELEMENT** has two *attributes*: *Count* and *Point Prefix*.

The **LCNMapping ELEMENT** is empty, meaning it has no sub-**ELEMENT**s.

The **LCNMapping ELEMENT** has two *attributes*: *Prefix* and *SurrogateServer*.

A *#REQUIRED attribute* must have a value, an *#IMPLIED attribute* does not need a value.

XML Environment Definition

Below is an example of XML vocabulary defining Data Access Redirection.

```
<?xml version="1.0"?>
<!--CONFIGURATION ITEM: UNC of the Environment Definition file must be
specified below-->
<!DOCTYPE Environment SYSTEM "FILE:///Program
Files/Honeywell/TPS/WebGUS/EnvDTDExample.dtd.xml">

<Environment FileSource="C:\DesktopGUS\Environments\MidlandEnv.xml">
  <DARedirection>
    <Honeywell>
      <LCNMappings Count="3" PointPrefix="G5\ ">
        <LCNMapping Prefix="lcn.G1\"
          SurrogateServer="AMWB2_TPNServer"/>
        <LCNMapping Prefix="lcn.G2\"
          SurrogateServer="AMWTB2_PHDServer"/>
        <LCNMapping Prefix="lcn.G5\"
          SurrogateServer="AMWB2_TPNServer"/>
      </LCNMappings>
    </Honeywell>
  </DARedirection>
```

Example of OPCDA server in XML

XML Environment DTD

Below is the Document Type Definition for OPCDA Server XML vocabulary.

```
<!-- OPCDAServers Section -->
<!ELEMENT OPCDAServers (OPCDAServer*)>
  <!--ATTLIST OPCDAServers Count CDATA #REQUIRED-->
  <!--ELEMENT OPCDAServer EMPTY -->
  <!--ATTLIST OPCDAServer  CollectionRateOverride  CDATA #IMPLIED
                           Description              CDATA #IMPLIED
                           Host                    CDATA #REQUIRED
                           Name                    CDATA #REQUIRED
                           PingParameter           CDATA #IMPLIED
                           ProgID                  CDATA #REQUIRED -->
```

Where:

The **OPCDAServers** **ELEMENT** has multiple **ELEMENT**s, each one an **OPCDAServer** **ELEMENT**.

The **OPCDAServers** **ELEMENT** has one *attribute*: *Count*.

The **OPCDAServer** **ELEMENT** is empty.

The **OPCDAServer** **ELEMENT** has five *attributes*: *CollectionRateOverride*, *Description*, *Host*, *Name*, and *ProgID*

XML Environment Definition

Below is an example of XML vocabulary defining the OPCDA Servers.

```
<!-- OPCDAServers Section -->
<Environment>
  <OPCDAServers Count="3">
    <OPCDAServer CollectionRateOverride="10"
      Description="TPN Server for Unit 1"
      Host="xyz-aa2-S009DE"
      Name="AMWB2_TPNServer"
      ProgID="Hci.TPNServer" />
    <OPCDAServer CollectionRateOverride="60"
      Description="Plant History Data for Unit 10"
      Host="systest57"
      Name="AMWTB2_PHDServer"
      ProgID="Hci.PHD"/>
    <OPCDAServer
      Description="an OPC DA server"
      Host="wi-xyz-bcm010"
      Name="AMPLC_OPCTServer"
      ProgID="{ADF6AE8B-B0F1-11d0-8A01-00C04FC97D9D}" />
  </OPCDAServers>
</Environment>
```

Example of Runtime parameters in XML

XML Environment DTD

Below is the Document Type Definition for Runtime parameter vocabulary.

```
<!-- RuntimeParameters Section -->
<!ELEMENT RuntimeParameters EMPTY >
  <!ATTLIST RuntimeParameters AccessTimeout CDATA #IMPLIED
                                AllowUnvalidatedDisplays CDATA #IMPLIED
                                BindTimeout CDATA #IMPLIED
                                CursorChangeResponseTimeout CDATA
#IMPLIED
                                DataPeriodicTimeout CDATA #IMPLIED
                                PCTSearchPath CDATA #IMPLIED
                                PingTimeout CDATA #IMPLIED
                                ShutdownWaitLimitTimeout CDATA #IMPLIED
                                UIExecuteTimeout CDATA #IMPLIED >
```

Where;

The **RuntimeParameters ELEMENT** is empty.

The **RuntimeParameters ELEMENT** has eight attributes: *AccessTimeout*, *BindTimeout*, *CursorChangeResponseTimeout*, *DataPeriodicTimeout*, *PCTSearchPath*, *PingTimeout*, *ShutdownWaitLimitTimeout*, and *UIExecuteTimeout*.

XML Environment Definition

Below is an example of XML vocabulary defining the Runtime Parameters.

```
<!-- RuntimeParameters Section -->
<Environment>
  <RuntimeParameters AccessTimeout="10000"
                    BindTimeout="5000"
                    CursorChangeResponseTimeout="500"
                    DataPeriodicTimeout="0"
                    PCTSearchPath="C:\localDisplays;u:\commondisplays;//re
f-tx15-s051/Honeywell/Displays"
                    PingTimeout="30000"
                    ShutdownWaitLimitTimeout="10000"
                    UIExecuteTimeout="0"/>
</Environment>
```

Consolidated DTD

Below is a sample of the three sections described above combined into one DTD file.

```
<!ELEMENT Environment (DARedirection, OPCDAServers, RuntimeParameters)>
<!-- Attribute for Environment -->
<!ATTLIST Environment FileSource          CDATA #REQUIRED>
<!-- DARedirection Section -->
<!ELEMENT DARedirection (Honeywell)>
  <!ELEMENT Honeywell (LCNMappings)>
    <!ELEMENT LCNMappings ( LCNMapping* )>
      <!ATTLIST LCNMappings Count          CDATA #REQUIRED
                          PointPrefix      CDATA #IMPLIED >
    <!ELEMENT LCNMapping
      <!ATTLIST LCNMapping Prefix          CDATA #IMPLIED >
                          SurrogateServer CDATA #REQUIRED>

<!-- OPCDAServers Section -->
<!ELEMENT OPCDAServers (OPCDAServer*)>
  <!ATTLIST OPCDAServers Count          CDATA #REQUIRED>
  <!ELEMENT OPCDAServer
    <!ATTLIST OPCDAServer CollectionRateOverride CDATA
#IMPLIED
                          Description      CDATA #IMPLIED
                          Host              CDATA #REQUIRED
                          Name              CDATA #REQUIRED
                          ProgID           CDATA #REQUIRED >

<!-- RuntimeParameters Section -->
<!ELEMENT RuntimeParameters          EMPTY >
  <!ATTLIST RuntimeParameters AccessTimeout CDATA #IMPLIED
                          BindTimeout        CDATA #IMPLIED
                          CursorChangeResponseTimeout CDATA #IMPLIED
                          DataPeriodicTimeout CDATA #IMPLIED
                          PCTSearchPath      CDATA #IMPLIED
                          PingTimeout        CDATA #IMPLIED
                          ShutdownWaitLimitTimeout CDATA #IMPLIED
                          UIExecuteTimeout   CDATA #IMPLIED
```

7 WebGUS Environment XML Data Files

7.1 XML Detailed Description

Honeywell

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Phoenix, AZ 85027 USA